CENG7880
Trustworthy and Responsible AI

Instructor: Sinan Kalkan

(https://ceng.metu.edu.tr/~skalkan)

For course logistics and materials:
https://metu-trai.github.io



MIDDLE EAST TECHNICAL UNIVERSITY
DEPARTMENT OF COMPUTER ENGINEERING



Weekly outline

Previously on CENG7880!

Week & Date		Topic
1	28 Sep	**Course logistics and overview** [Intro to trust and responsibility in AI: Social, ethical and policy implications of AI]
2	5 Oct	**Overview** [Deep/machine learning concepts (fundamentals)]
3	12 Oct	**Overview** [Deep/machine learning concepts (recent trends)]
4	19 Oct	**Robust AI** [Fundamentals of Robust AI]
5	26 Oct	**Robust AI** [Adversarial Attacks & Corruptions]
6	2 Nov	**Robust AI** [Defense Mechanisms and Robust Training]
7	9 Nov	**Robust AI** [Quantifying Uncertainty and Calibration of Deep Networks]
8	16 Nov	**Explainable AI** [Fundamentals of Explainable AI]
9	23 Nov	**Explainable AI** [Model-specific Explainability Methods: Explainable shallow models, prototype-based models, concept bottleneck models, attention mechanisms for explainability]
10	30 Nov	**Explainable AI** [Post-Hoc Explanation Techniques: CAM, GradCAM, feature-based saliency estimation, counterfactual reasoning]
11	7 Dec	**Explainable AI** [Evaluating explainability, methods and metrics]
12	14 Dec	**Bias and Fairness in AI** [Sources of bias; definitions of fairness; measures of fairness]
13	21 Dec	**Bias and Fairness in AI** [Data-level and in-processing methods for fairness]
14	28 Dec	**Bias and Fairness in AI** [Regularization-based and post-processing methods for fairness]

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Grading

Grading:

Quizzes	20%
Project	35%
Final Exam	45%

Warnings

Participating in the final exam is subject to the following conditions: (i) Attending 60% of the quizzes. (ii) Completing half of the project. Not satisfying these will result in NA as the grade.

Previously on CENG7880!

Project

- Reproduce a conference paper without any implementation
  - Reproduce their results and provide a critical analysis
  - Produce a Github Repo with a detailed Readme file
- Papers can **only** be from the following top conferences:
  - ICLR, AAAI, NeurIPS, ICML, CVPR, ECCV or similar
- You can work in groups of two.
- Form for submitting chosen papers
  - Form: https://forms.gle/Z8SFya9hQtdzP7jG6
  - Deadline for choosing papers: 16 October.

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Principles of Trust in AI

1. Explainability
2. Fairness/Justice
3. Privacy
4. Robustness
5. Accountability
6. Autonomy
7. Transparency
8. Beneficience
9. Maleficience

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Robustness

- "ML Model robustness denotes the capacity of a model to sustain stable predictive performance in the face of **variations and changes in the input data**"

https://arxiv.org/pdf/2404.00897v2

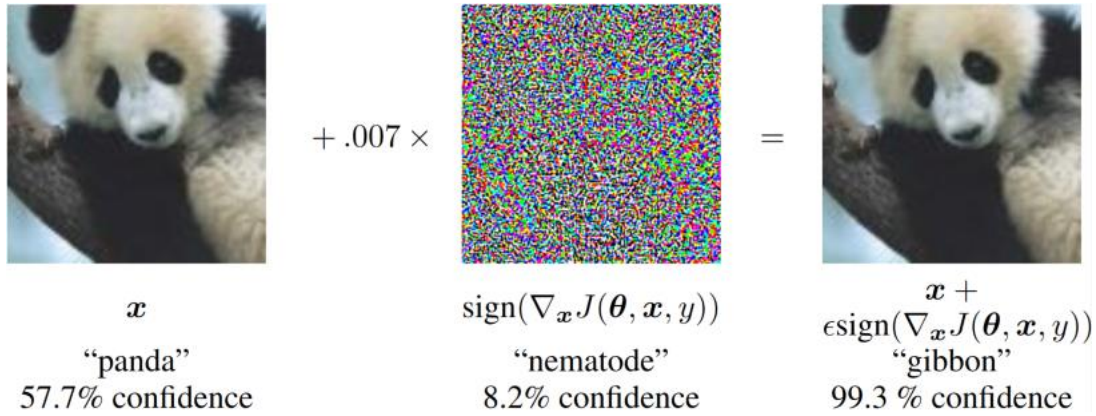
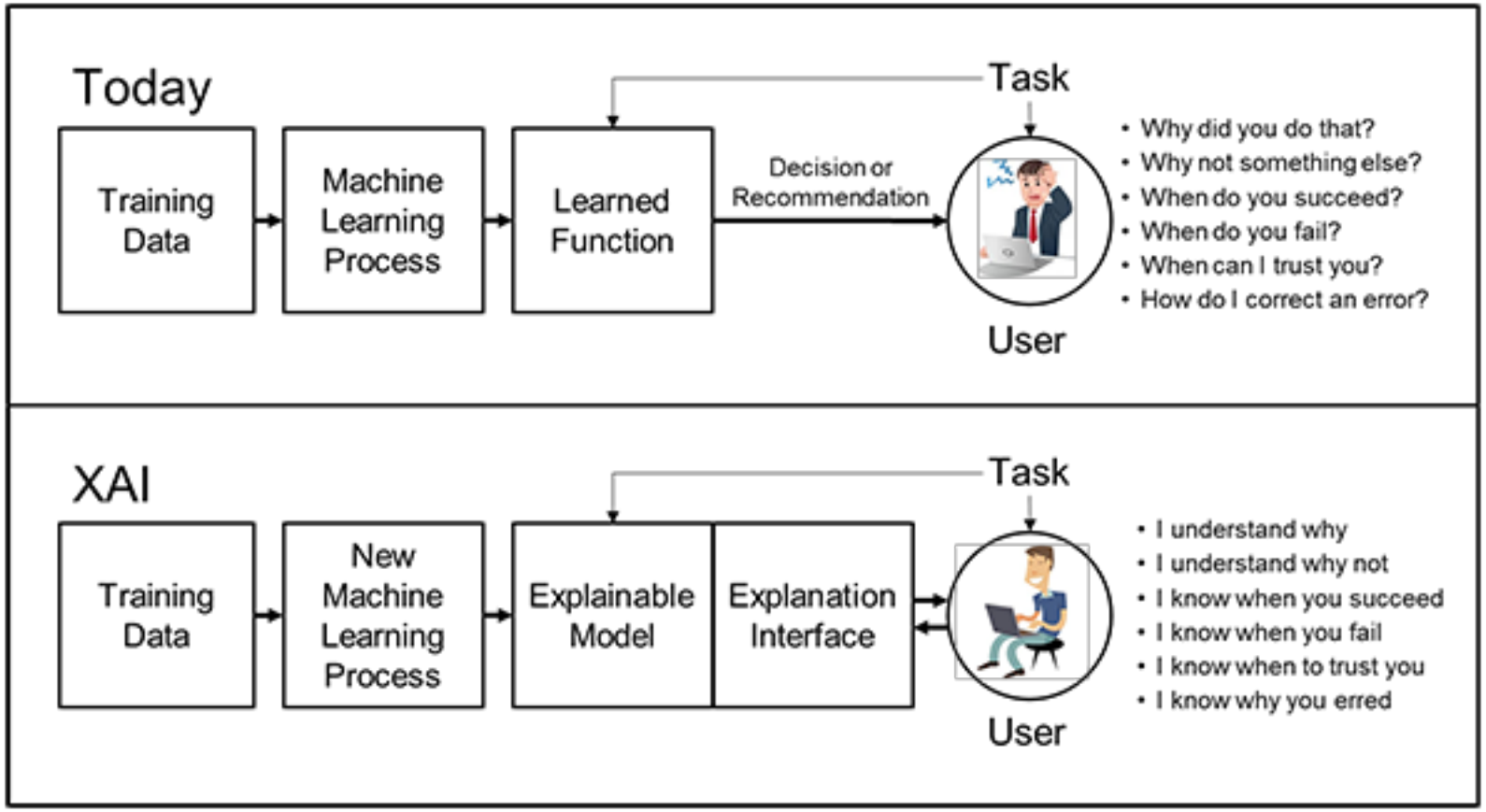


Fig: https://arxiv.org/pdf/1412.6572

Previously on CENG7880!

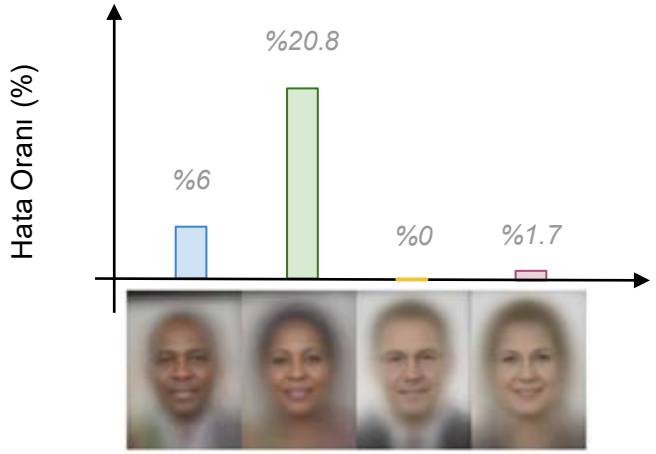
Explainability



https://www.darpa.mil/research/programs/explainable-artificial-intelligence

Previously on CENG7880!

Bias and Fairness



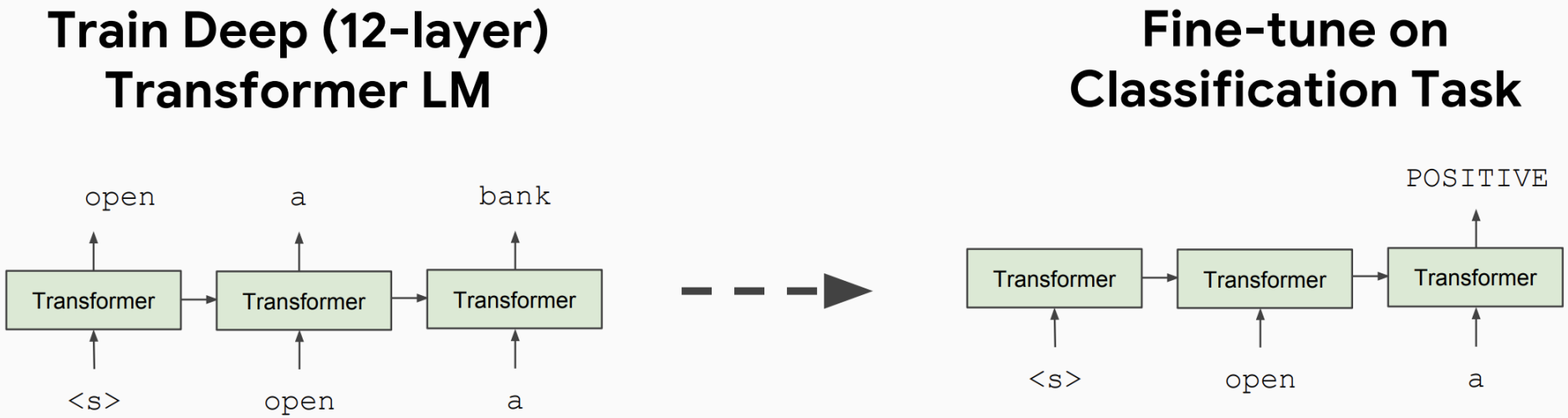
Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%

Buolamwini, J., Gebru, T.: Gender shades: Intersectional accuracy disparities in commercial gender classification. In: Conference on fairness, accountability and transparency. pp. 77-91 (2018).

Previously on CENG7880!

Pre-training in NLP

- *Improving Language Understanding by Generative Pre-Training*, OpenAI, 2018



Slide: Jacob Devlin
https://nlp.stanford.edu/seminar/details/jdevlin.pdf

Previously on CENG7880!

GPT-1

- 12 layer decoder-only transformer
- Unsupervised pretraining
  - BookCorpus dataset
- Supervised finetuning
  - Textual alignment
  - QA & commonsense reasoning
  - Semantic similarity
  - Classification

OpenAI Technical Report, 2018

Improving Language Understanding by Generative Pre-Training

Alec Radford, OpenAI, alec@openai.com
Karthik Narasimhan, OpenAI, karthikn@openai.com
Tim Salimans, OpenAI, tim@openai.com
Ilya Sutskever, OpenAI, ilyasu@openai.com

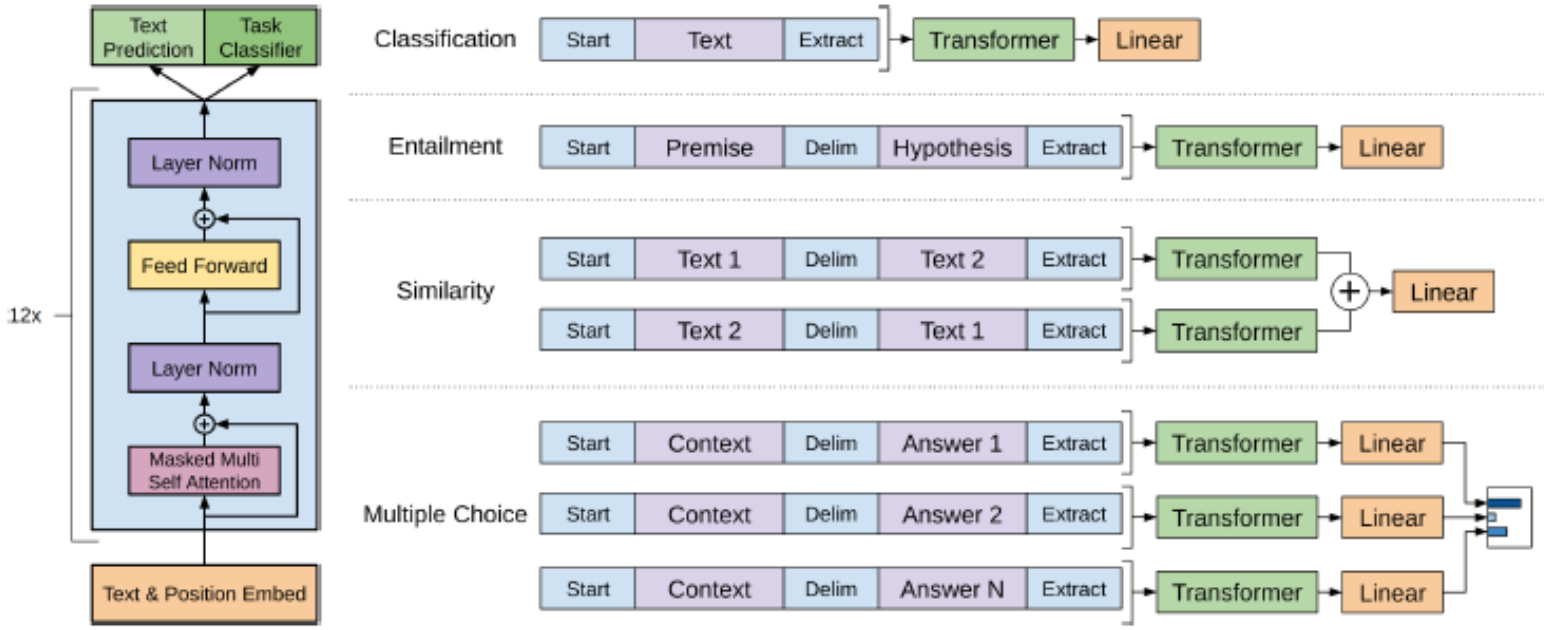


Figure 1: **(left)** Transformer architecture and training objectives used in this work. **(right)** Input transformations for fine-tuning on different tasks. We convert all structured inputs into token sequences to be processed by our pre-trained model, followed by a linear+softmax layer.

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GPT-2

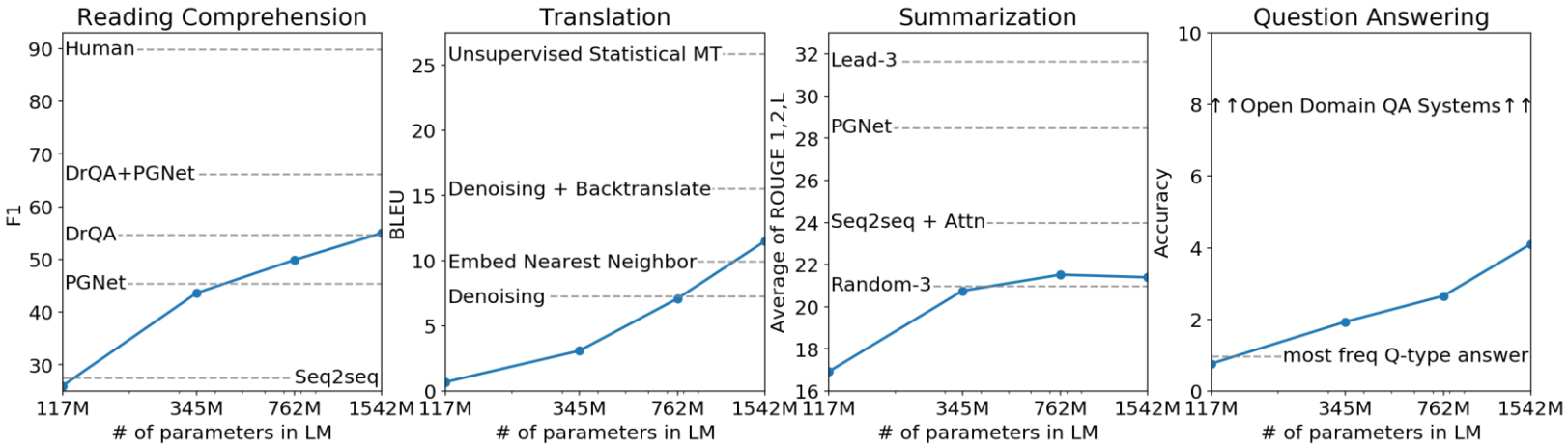
OpenAI Technical Report, 2019.

Language Models are Unsupervised Multitask Learners

Alec Radford [*1] Jeffrey Wu [*1] Rewon Child [1] David Luan [1] Dario Amodei [**1] Ilya Sutskever [**1]

Abstract

Natural language processing tasks, such as question answering, machine translation, reading comprehension, and summarization, are typically approached with supervised learning on task-specific datasets. We demonstrate that language models begin to learn these tasks without any explicit supervision when trained on a new dataset of millions of webpages called WebText. When conditioned on a document plus questions, the answers generated by the language model reach 55 F1 on the CoQA dataset - matching or exceeding the performance of 3 out of 4 baseline systems without using the 127,000+ training examples. The capacity of the language model is essential to the success of zero-shot task transfer and increasing it improves performance in a log-linear fashion across tasks. Our largest model, GPT-2, is a 1.5B parameter Transformer that achieves state of the art results on 7 out of 8 tested language modeling datasets in a zero-shot setting but still underfits WebText. Samples from the model reflect these improvements and contain coherent paragraphs of text. These findings suggest a promising path towards building language processing systems which learn to perform tasks from their naturally occurring demonstrations.



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GPT-3

- 175B parameters!

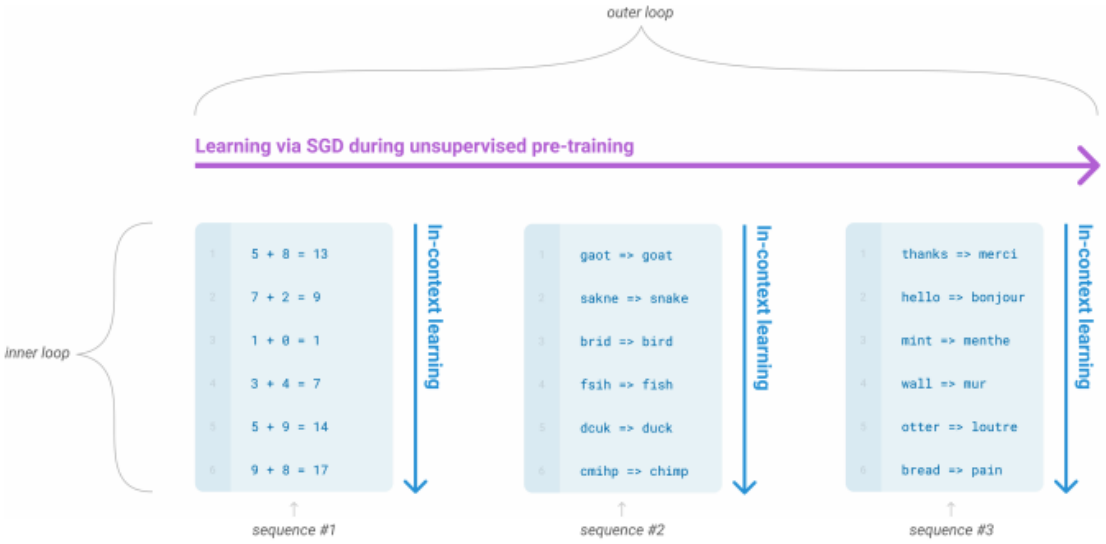


Figure 1.1: Language model meta-learning. During unsupervised pre-training, a language model develops a broad set of skills and pattern recognition abilities. It then uses these abilities at inference time to rapidly adapt to or recognize the desired task. We use the term "in-context learning" to describe the inner loop of this process, which occurs within the forward-pass upon each sequence. The sequences in this diagram are not intended to be representative of the data a model would see during pre-training, but are intended to show that there are sometimes repeated sub-tasks embedded within a single sequence.

Language Models are Few-Shot Learners

Tom B. Brown* Benjamin Mann* Nick Ryder* Melanie Subbiah*

Jared Kaplan† Prafulla Dhariwal Arvind Neelakantan Pranav Shyam Girish Sastry

Amanda Askell Sandhini Agarwal Ariel Herbert-Voss Gretchen Krueger Tom Henighan

Rewon Child Aditya Ramesh Daniel M. Ziegler Jeffrey Wu Clemens Winter

Christopher Hesse Mark Chen Eric Sigler Mateusz Litwin Scott Gray

Benjamin Chess Jack Clark Christopher Berner

Sam McCandlish Alec Radford Ilya Sutskever Dario Amodei

OpenAI

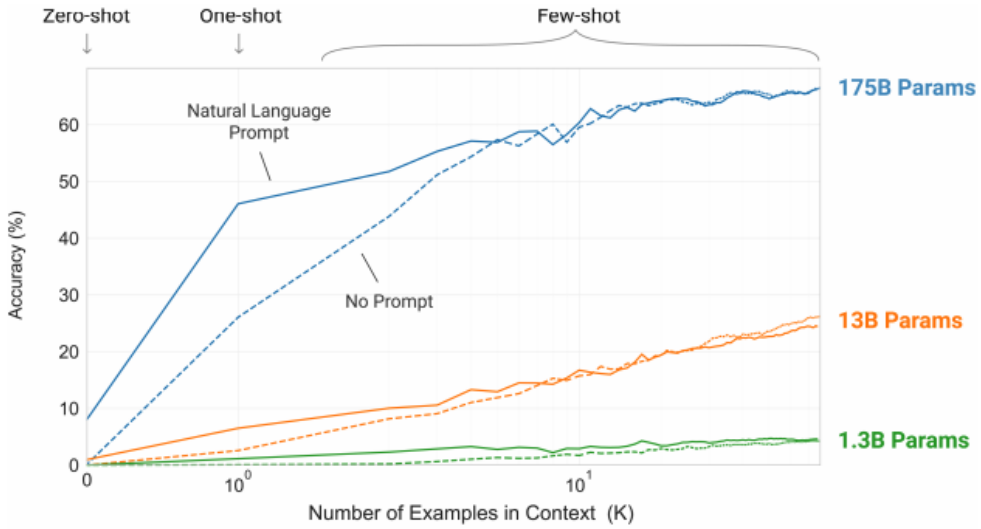


Figure 1.2: Larger models make increasingly efficient use of in-context information. We show in-context learning performance on a simple task requiring the model to remove random symbols from a word, both with and without a natural language task description (see Sec. 3.9.2). The steeper "in-context learning curves" for large models demonstrate improved ability to learn a task from contextual information. We see qualitatively similar behavior across a wide range of tasks.

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GPT-3

Three ways of in-context learning:

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  cheese =>                           ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  sea otter => loutre de mer          ← example
3  cheese =>                           ← prompt
```

In a single sequence input, the prompted example can learn from previous demonstrations.

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  sea otter => loutre de mer          ← examples
3  peppermint => menthe poivrée        ←
4  plush girafe => girafe peluche      ←
5  cheese =>                           ← prompt
```

Slide: Weizhi Wang
https://victorwz.github.io/additional_files/slides_gpt_cs291A.pdf

Previously on CENG7880!

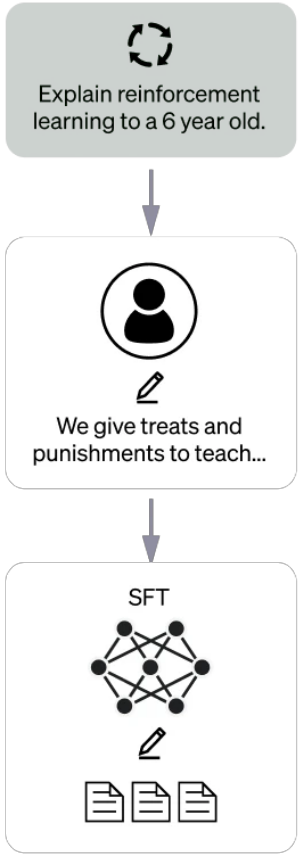
Step 1

Collect demonstration data and train a supervised policy.

A prompt is sample from our prompt dataset.

A labeler demonstrates the desired output behavior.

This data is used to fine-tune GPT-3.5 with supervised learning.



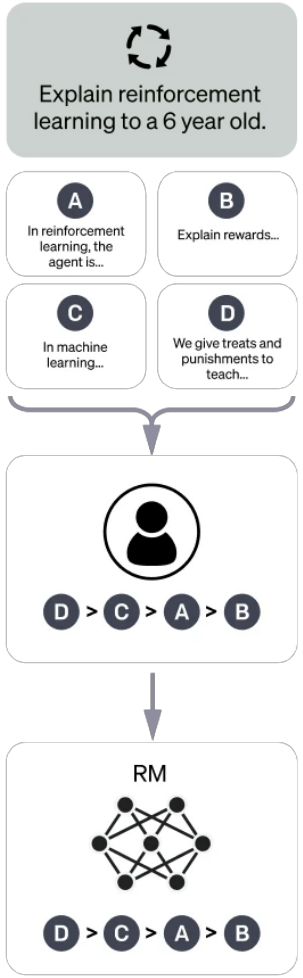
Step 2

Collect comparison data and train a reward model.

A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.

This data is used to train our reward model.



Step 3

Optimize a policy against the reward model using the PPO reinforcement learning algorithm.

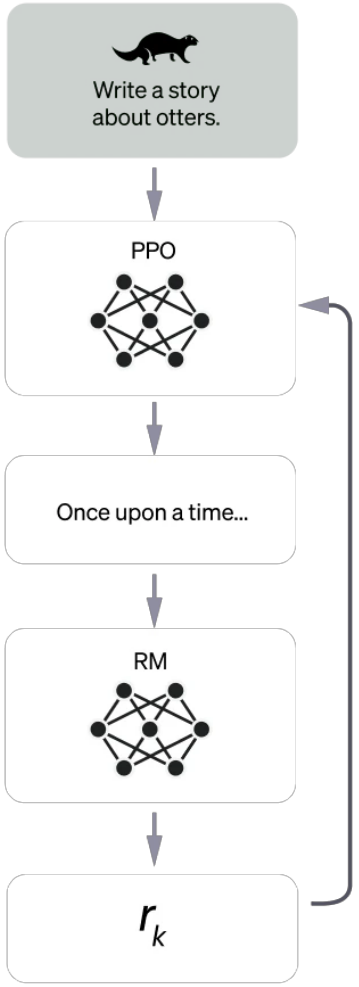
A new prompt is sampled from the dataset.

The PPO model is initialized from the supervised policy.

The policy generates an output.

The reward model calculates a reward for the output.

The reward is used to update the policy using PPO.



https://openai.com/index/chatgpt/

Reading of the Week

What if automating AI R&D triggers an intelligence explosion?

Alan Chan*	GovAI
Christoph Winter	CASP, University of Cambridge, Institute for Law & AI
Andrew Barto	University of Massachusetts Amherst
Jakub Pachocki	OpenAI
Geoffrey Hinton	University of Toronto, Vector Institute
Eric Horvitz	Microsoft
Yoshua Bengio	Mila (Quebec AI Institute), Université de Montréal, LawZero
Dawn Song	University of California, Berkeley
Jack Clark	Anthropic
Hilary Greaves	University of Oxford
Anton Korinek	University of Virginia, Anthropic
Samuel Hammond	Foundation for American Innovation
Thore Graepel	University College London
Ben Bariach	University of Oxford
Philip H. S. Torr	University of Oxford
Sheila A. McIlraith	University of Toronto, Vector Institute
Jeff Clune	University of British Columbia, Vector Institute
Sam Manning	GovAI, Foundation for American Innovation
Girish Sastry	Guidelight
Tom Davidson	Forethought
Daniel Eth	AI Policy Institute
Sören Mindermann*	CASP, University of Cambridge

Abstract

In contrast to even a year ago, AI systems now write most of the code inside the companies that build them. As more of the AI research and development (R&D) pipeline is automated, could AI progress radically accelerate in an "intelligence explosion," where years of advances are compressed into months or less? Preliminary evidence suggests that it could. In this work, we assess this evidence, analyze an intelligence explosion's potential impacts, and propose policy responses. AI systems are on track to automate most AI R&D work within a few years, and possibly all of it. If this triggers an intelligence explosion, it could dramatically bring forward AI's benefits, but also pose extreme risks: capabilities growth could accelerate far beyond what society can keep up with, humanity could lose control over superhuman AI systems, and checks on power within and between states, companies, and branches of government could be severely eroded. Although there remains much uncertainty about these possibilities, the high stakes warrant serious further attention. Policymakers should urgently obtain more visibility into the automation of AI R&D, develop ways to steer and constrain an intelligence explosion, and prepare society to adapt to an intelligence explosion's impacts.

28 September 2026

Agenda

- Robustness
  - Distribution shifts (this week)
  - Adversarial robustness (next week)
  - Certifying robustness (next next week)
  - Calibrated predictions & uncertainty (next next next week)

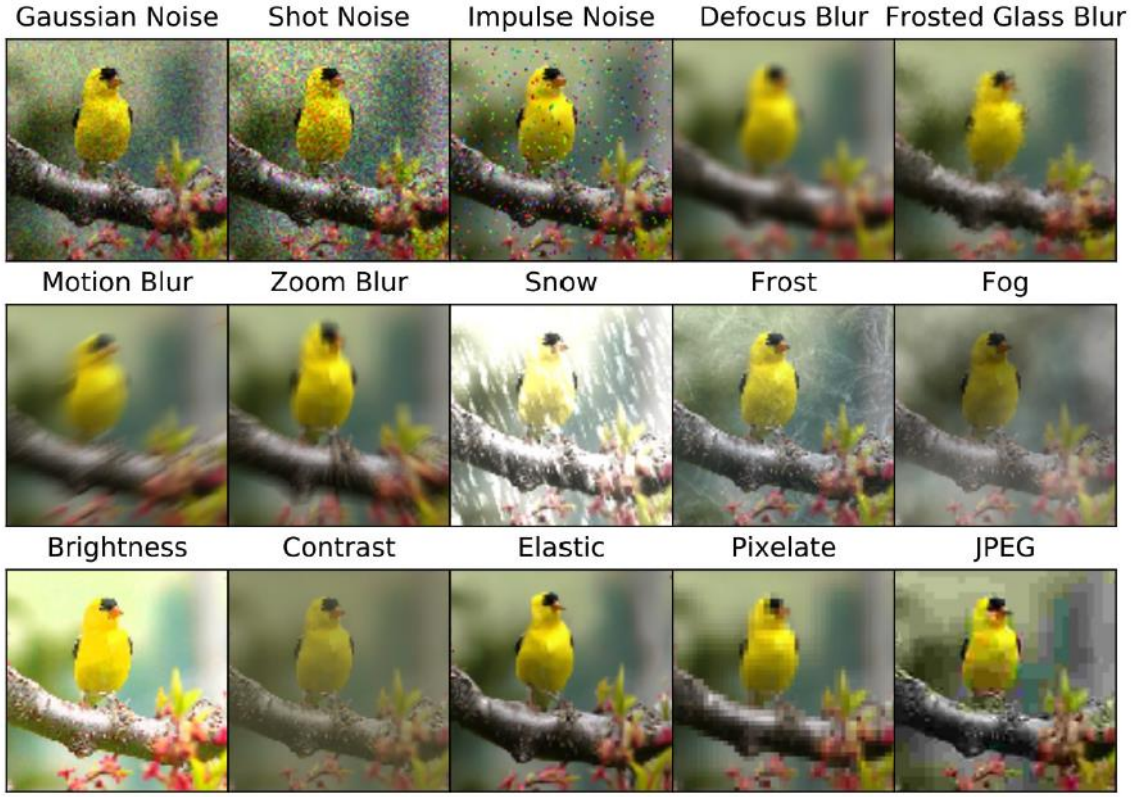
Administrative Notes

- Registration
  - Everyone registered?
  - Everyone has access to ODTUclass

- Selecting papers
  - Form: https://forms.gle/Z8SFya9hQtdzP7jG6
  - Deadline for choosing papers: 16 October.

Robustness: Agenda
Robustness to Distribution Shifts

- Images
  - Noise, color changes, light changes, day/night, summer/winter, …
- Audio
  - Noise, background, gain level, …
- Text
  - Noise, synonyms, alternative phrasing, ..

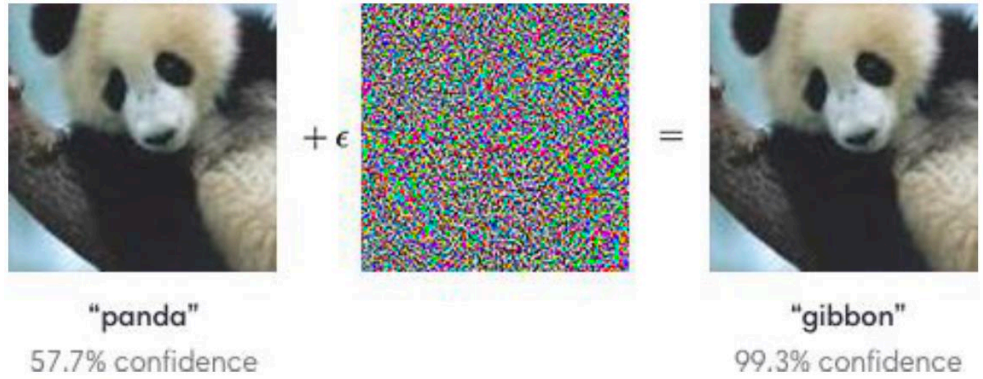


Hendrycks & Dietterich, Benchmarking Neural Network Robustness to Common Corruptions and Perturbations

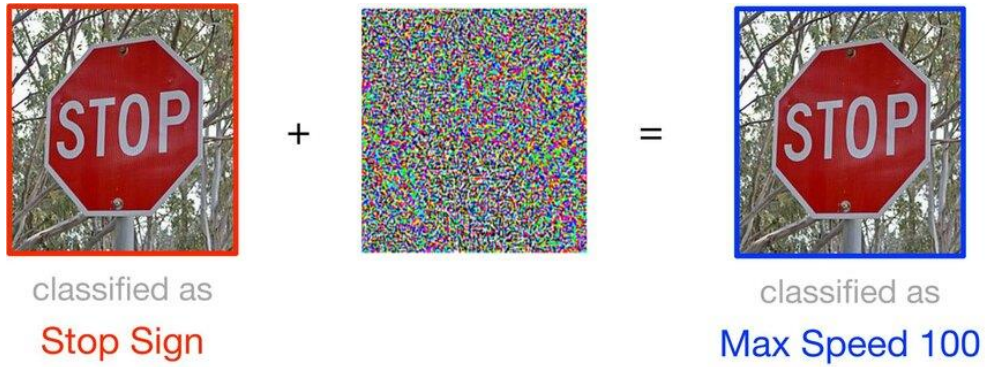
Robustness: Agenda
Adversarial Robustness

- Types of adversarial attacks

- Adversarial training



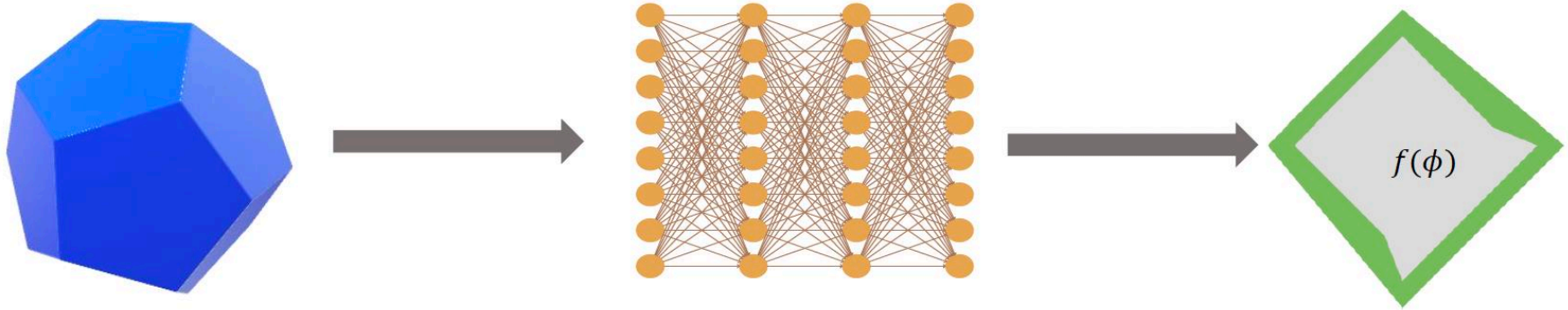
Szegedy et al., Intriguing Properties of Neural Networks, 2014



https://www.altacognita.com/robust-attribution/

Robustness: Agenda
Certifying/Verifying Robustness

Neural network certification: problem statement



Figs: Uni of Pennsylvania, Trustworthy ML - CIS 7000

Robustness: Agenda
Calibrated Predictions & Uncertainty

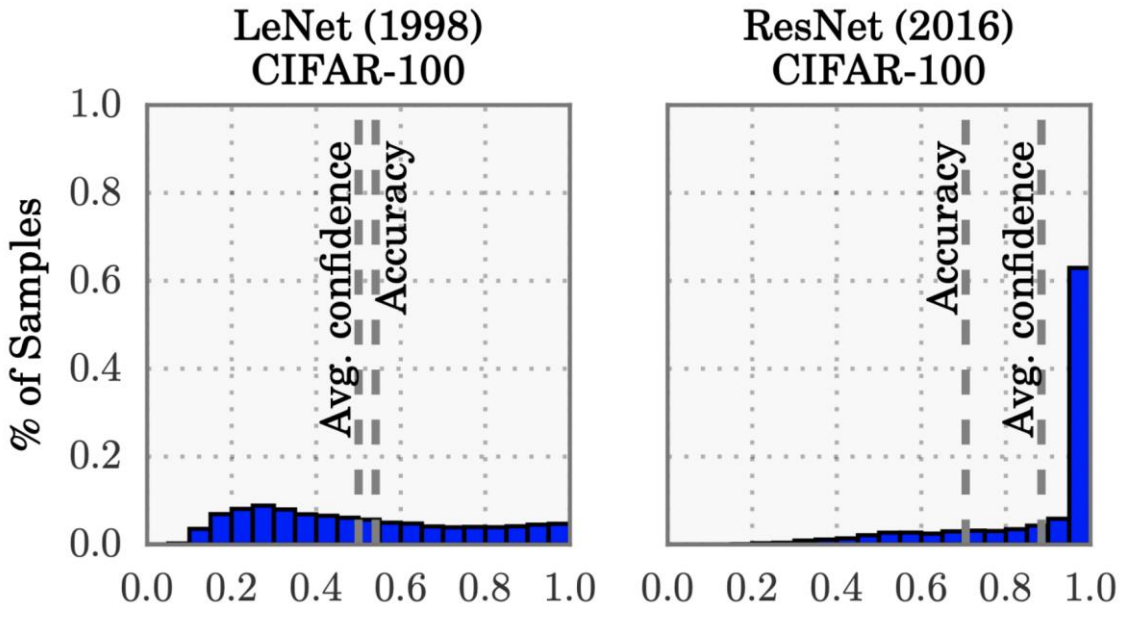


Fig: Uni of Pennsylvania, Trustworthy ML - CIS 7000

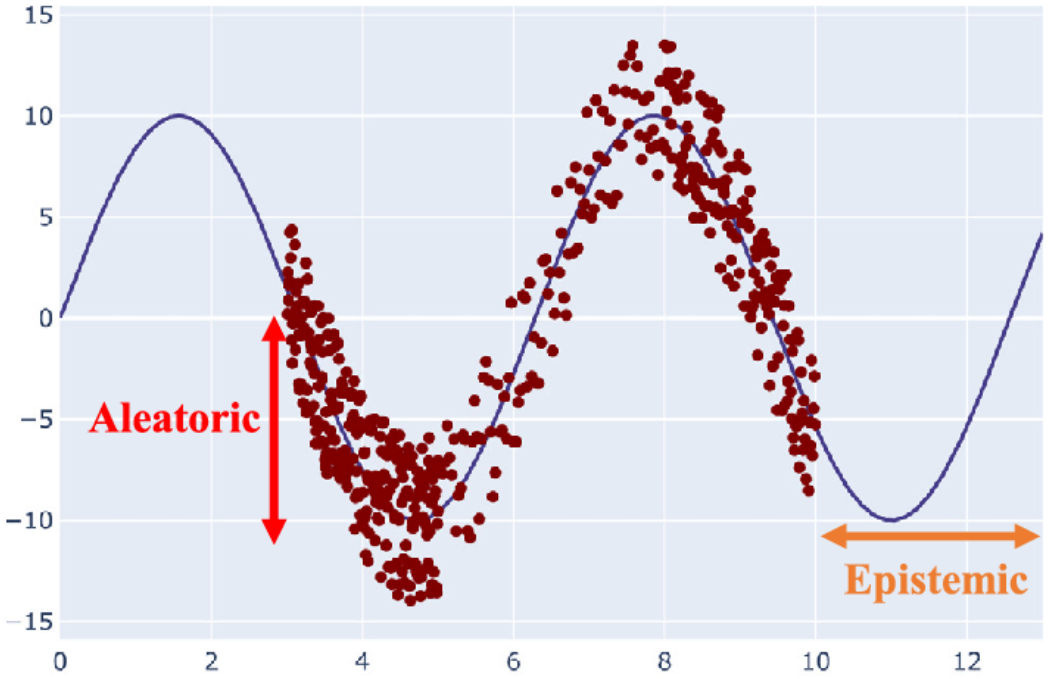


Fig: Abdar et al., A review of uncertainty quantification in deep learning: Techniques, applications and challenges, 2021

Background: Bayes' Theorem

- For events A and B:

conditional probability of event B occurring given event A is true. (likelihood)

$$p(A|B) = \frac{p(A,B)}{p(B)} = \frac{p(B|A)p(A)}{p(B)}$$

prior on A

prior on B (evidence)

conditional probability of event A occurring given event B is true. (posterior probability)

Background: Marginalization

- Marginal distribution: The distribution of some variables without reference/condition on other variables.
- Example:
  - X, Y: Two discrete random variables
  - Marginal distribution of X:

$$p(x_i) = \sum_j p(x_i, y_j) = \sum_j p(x_i \mid y_j) p(y_j)$$

  - Marginal distribution of Y:

$$p(y_i) = \sum_j p(x_j, y_i) = \sum_j p(y_i \mid x_j) p(x_j)$$

  - These turn into integrals in continuous cases:

$$p(x) = \int_y p(x, y) \mathrm{d}y = \int_y p(x \mid y) p(y) \, \mathrm{d}y$$

$$p(y) = \int_x p(x, y) \mathrm{d}x = \int_x p(y \mid x) p(x) \, \mathrm{d}x$$

Y \ X	x_1	x_2	x_3	x_4	$p_Y(y)$ ↓
y_1	$\frac{4}{32}$	$\frac{2}{32}$	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{8}{32}$
y_2	$\frac{3}{32}$	$\frac{6}{32}$	$\frac{3}{32}$	$\frac{3}{32}$	$\frac{15}{32}$
y_3	$\frac{9}{32}$	0	0	0	$\frac{9}{32}$
$p_X(x)$ →	$\frac{16}{32}$	$\frac{8}{32}$	$\frac{4}{32}$	$\frac{4}{32}$	$\frac{32}{32}$

Table from Wikipedia

Background: Expected Value

- Expected value (also called expectation, mean, or first moment) of a variable X
- Notations: $E(X), E[X]$

$$E[X] = x_1 p_1 + x_2 p_2 + \cdots x_k p_k = \sum_i x_i p_i$$

$$E[X] = \int x p(x) dx$$

Background:
Independent and identically distributed (i.i.d.)

- *"a collection of random variables is **independent** and **identically distributed** (i.i.d., iid, or IID) if each random variable has the same probability distribution as the others and all are mutually independent."* – Wikipedia
  - **identically distributed**: When samples are taken from the distribution, there is no overall trend / change / shift / fluctuation in the distribution.
    - More formally: $F_X(x) = F_Y(x)$ for $F_X(x) = P(X \leq x)$ and $F_Y(y) = P(Y \leq y)$ being the cumulative distributions.
  - **independent**: Samples are not "connected" to other samples. I.e., knowing one sample does not say anything about other samples.
    - More formally: $p(A, B) = p(A) \cdot p(B)$.
- Synonym with "random variable"

Background:
Independent and identically distributed (i.i.d.)

**Example from [Wikipedia](): **

Toss a coin 10 times and write down the results into variables $A_1, \ldots, A_{10}$.

1. **Independent**: Each outcome A_i will not affect the other outcome A_j (for $i \neq j$ from 1 to 10), which means the variables $A_1, \ldots, A_{10}$ are independent of each other.
2. **Identically distributed**: Regardless of whether the coin is fair (with a probability of 1/2 for heads) or biased, as long as the same coin is used for each flip, the probability of getting heads remains consistent across all flips.

Background: Infimum (inf) and Supremum (sup)

- Infimum
  - Greatest lower bound

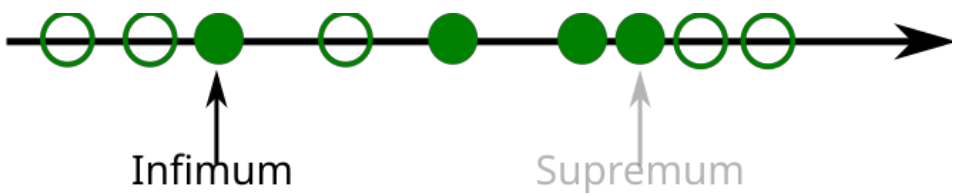


Fig: https://en.wikipedia.org/wiki/Infimum_and_supremum

- Supremum
  - Lowest upper bound

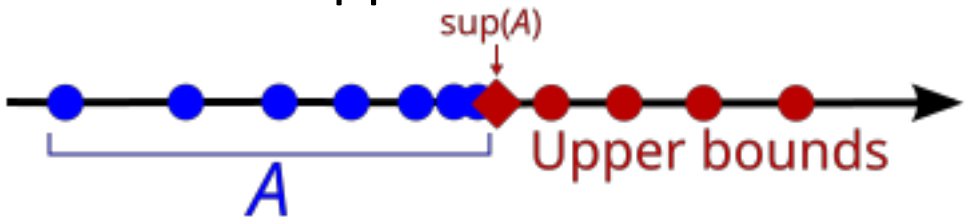


Fig: https://en.wikipedia.org/wiki/Infimum_and_supremum

Differences to min and max:

- inf and sup may not be part of the set

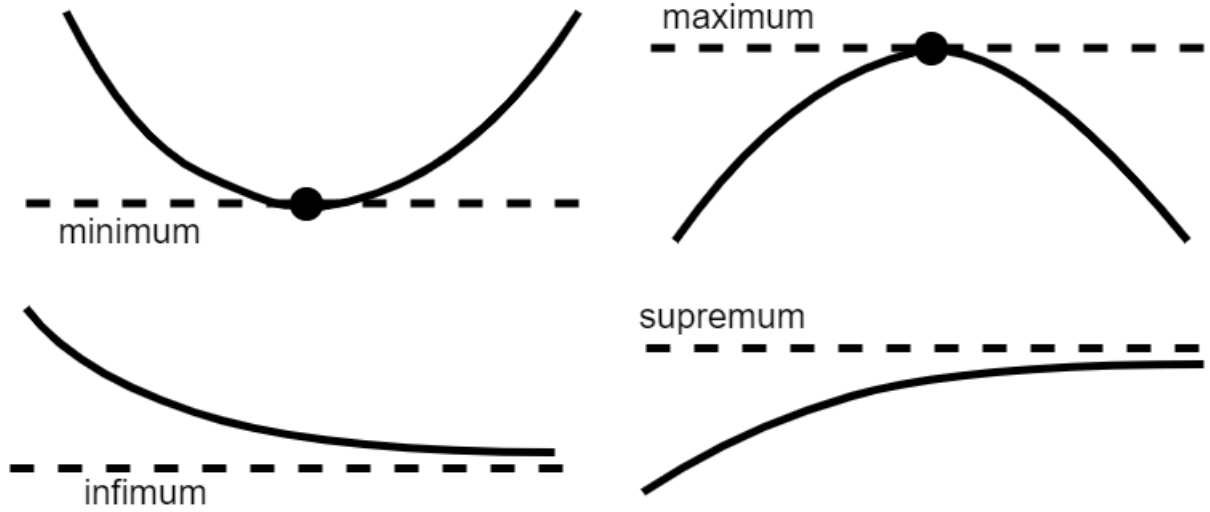


Fig: https://www.researchgate.net/figure/Minimum-maximum-infimum-and-supremum-of-example-functions_fig1_355093246

Background: Binomial Distribution

- If a random variable X follows the Binomial Distribution with parameter $n \in \mathbb{N}$ and $p \in [0, 1]$,
- then the probability of getting k successes (with same rate p) in n Bernoulli trials:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

with the binomial coefficient:

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

- $E[X] = np$
- $Var(X) = np(1-p)$

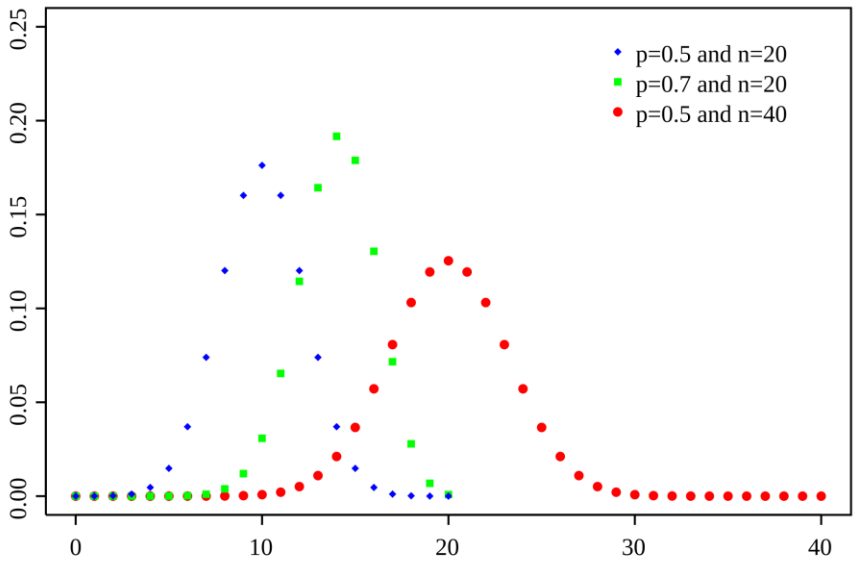


Fig: Wikipedia

Example from Wikipedia:

Suppose a biased coin comes up heads with probability 0.3 when tossed. The probability of seeing exactly 4 heads in 6 tosses is

$$f(4, 6, 0.3) = \binom{6}{4} 0.3^4 (1-0.3)^{6-4} = 0.059535.$$

Background: Bernoulli Distribution

- Discrete probability distribution where a variable can take two values
- X: a random Bernoulli variable, then:
  - $P(X = 1) = p$
  - $P(X = 0) = q = 1 - p$
  - $E[X] = p$
  - $Var[X] = p(1 - p)$
- A special case of Binomial Distribution with $n = 1$

Background: Lipschitz Constant

- For a function $f(\cdot)$:

$$K = \sup_{x_1 \neq x_2} \frac{|f(x_2) - f(x_1)|}{|x_2 - x_1|}$$

- Alternative definition:

$$|f(x_2) - f(x_1)| \leq K|x_2 - x_1|$$

- If the function is differentiable with bounded derivative:

$$K = \sup_{x} |f'(x)|$$

Background: Wasserstein Distance Metric

Also called: Kantorovich-Rubinstein metric, Earth Mover's Distance

- A measure between two probability distributions P and Q.
- The minimum "cost" of moving pile from P to Q.
  - A measure of optimal transport problem.
- For samples $X \sim P$ and $Y \sim Q$:

$$W(P,Q) = \left(\frac{1}{n}\sum_{i=1}^{n}\|X_i - Y_i\|^p\right)^{1/p}$$

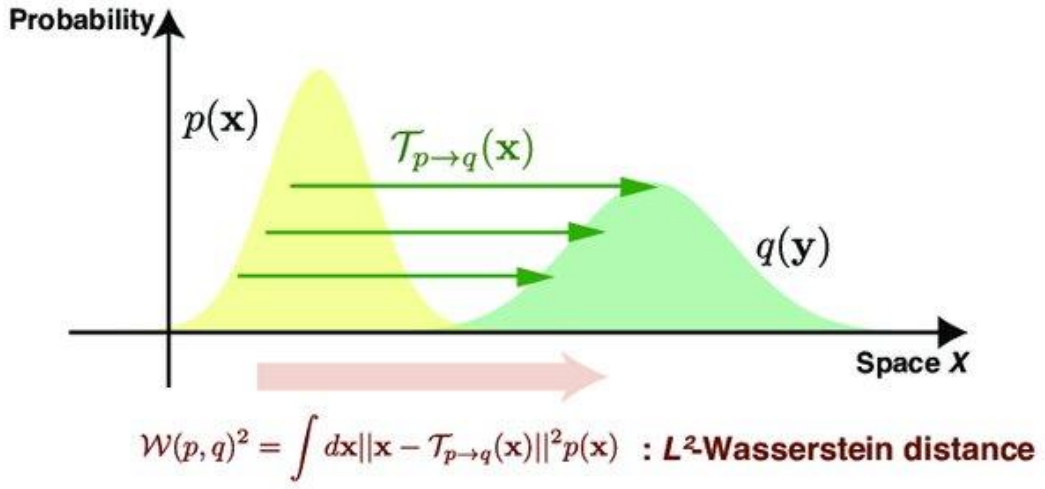


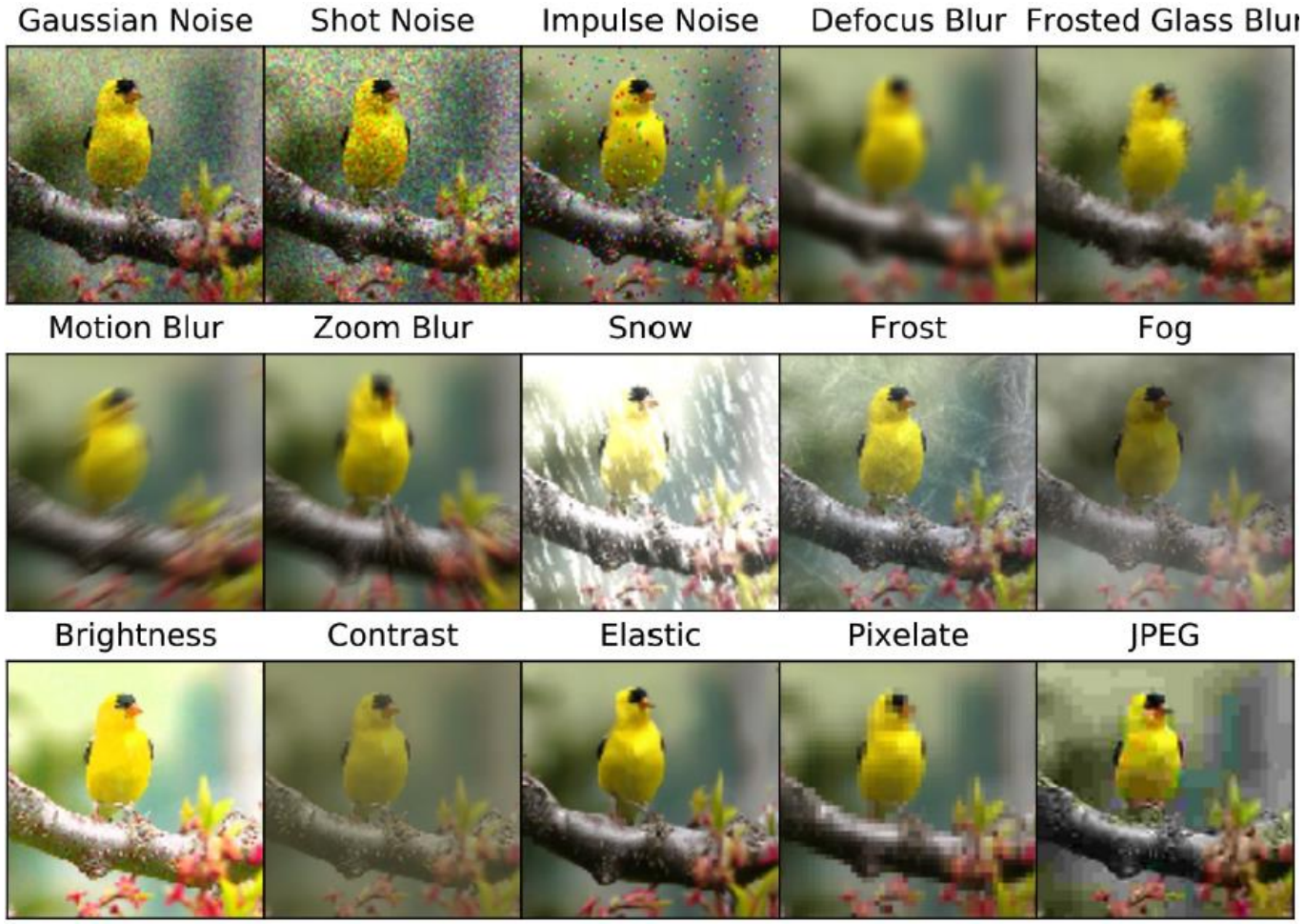
Fig: https://www.researchgate.net/publication/349704621_Geometrical_aspects_of_entropy_production_in_stochastic_thermodynamics_based_on_Wasserstein_distance/figures?lo=1

Robustness to Distribution Shift

Disclaimer: The slides are mostly borrowed from:

Uni of Pennsylvania, Trustworthy ML - CIS 7000

Example: Synthetic Perturbations



Hendrycks & Dietterich, Benchmarking Neural Network Robustness to Common Corruptions and Perturbations

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Example: Synthetic Perturbations

- **Significantly reduces performance**
  - 20% error rate → 80% error rate
- **Data augmentation can help (but not 100% solution)**

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Data Augmentation

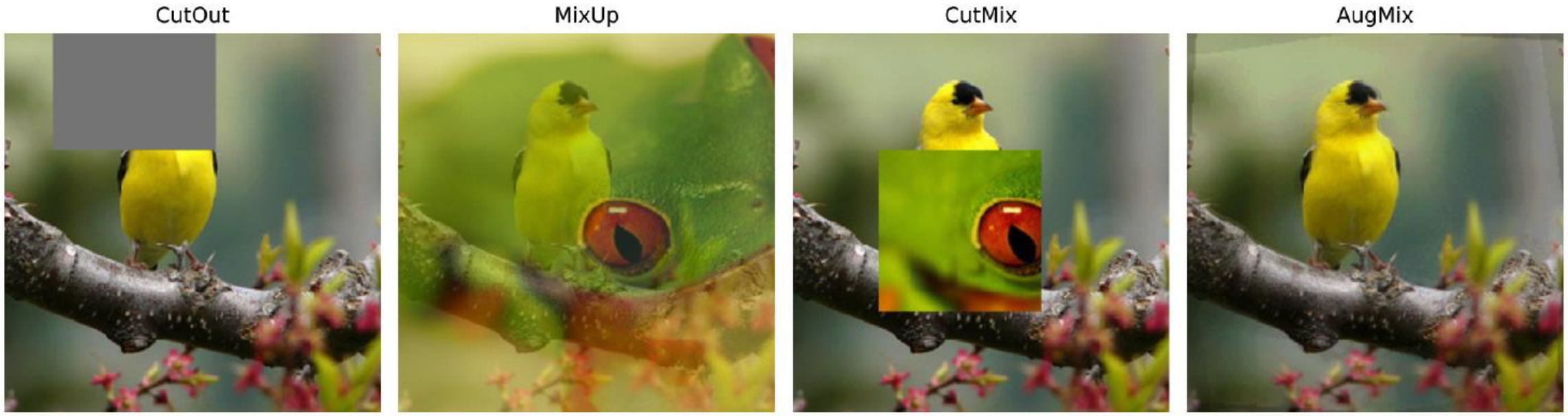


Figure 1: A visual comparison of data augmentation techniques. AugMix produces images with variety while preserving much of the image semantics and local statistics.

Hendrycks et al., AugMix: A Simple Data Processing Method to Improve Robustness and Uncertainty, 2020

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mixup: Beyond Empirical Risk Minimization, ICLR 2018.

MixUp

$$\begin{aligned} \hat{x} &= \lambda x_i + (1-\lambda) x_j, \\ \hat{y} &= \lambda y_i + (1-\lambda) y_j, \end{aligned}$$

where $\lambda \in [0, 1]$ is a random number

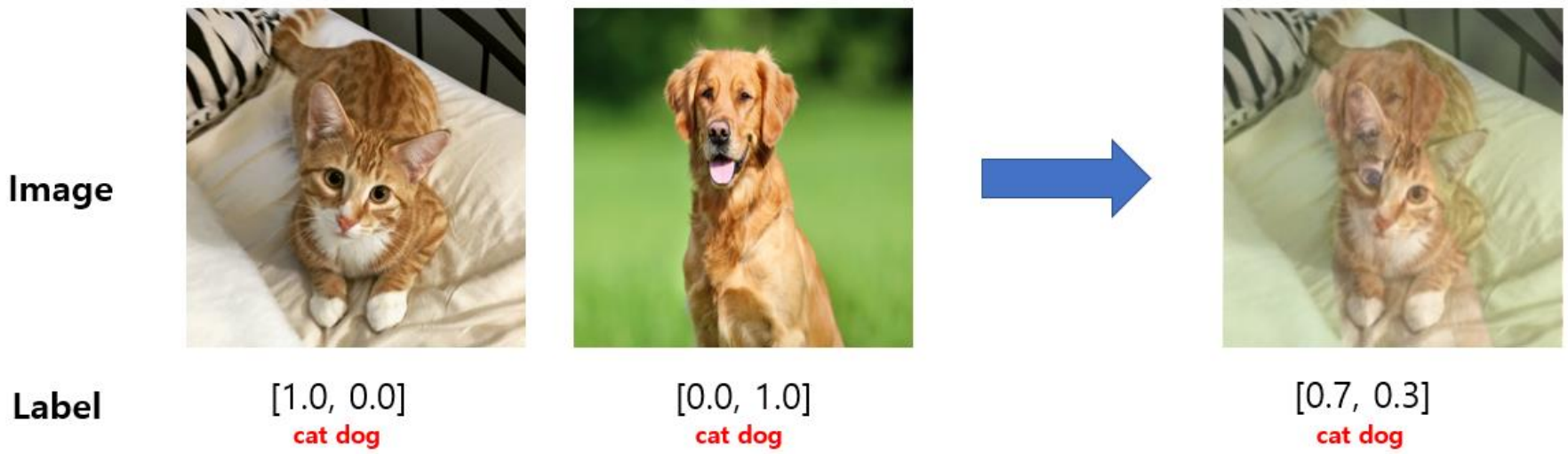


Fig: https://www.kaggle.com/code/kaushal2896/data-augmentation-tutorial-basic-cutout-mixup

AugMix

AugMix: A Simple Data Processing Method to Improve Robustness and Uncertainty, ICLR 2020.

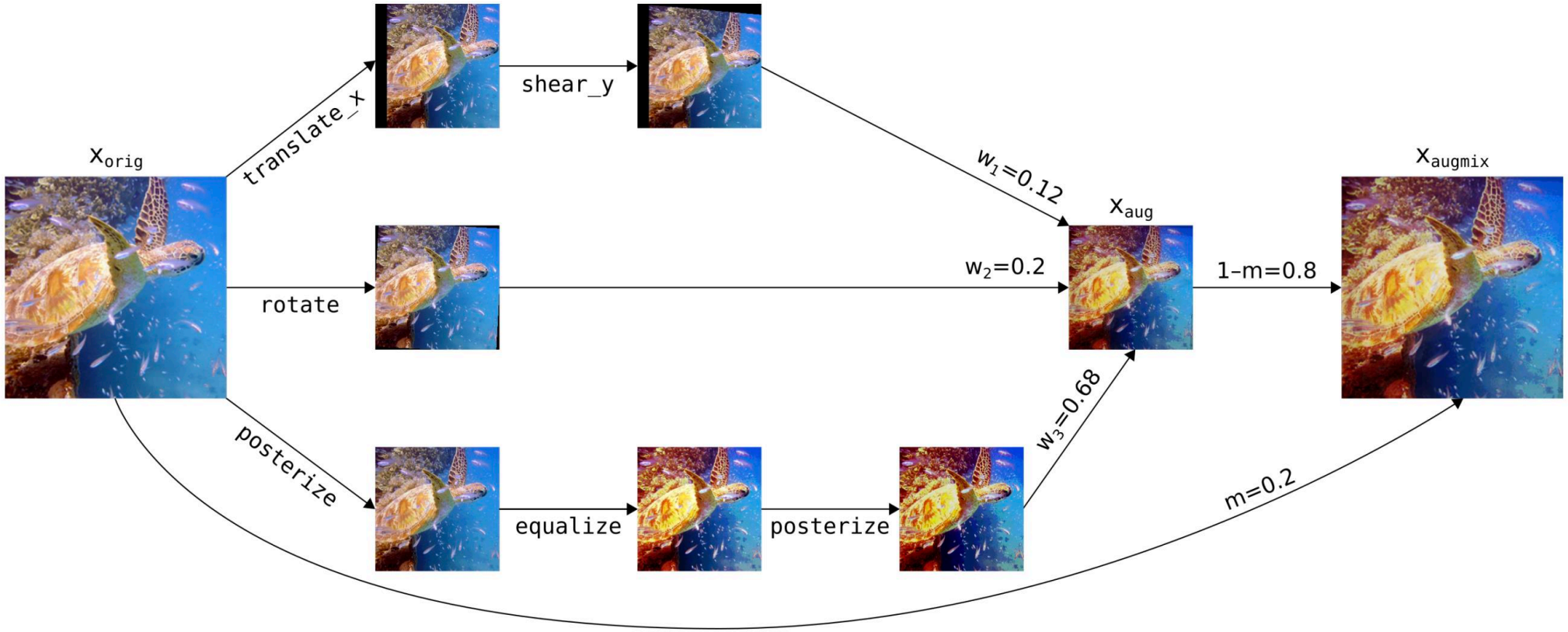
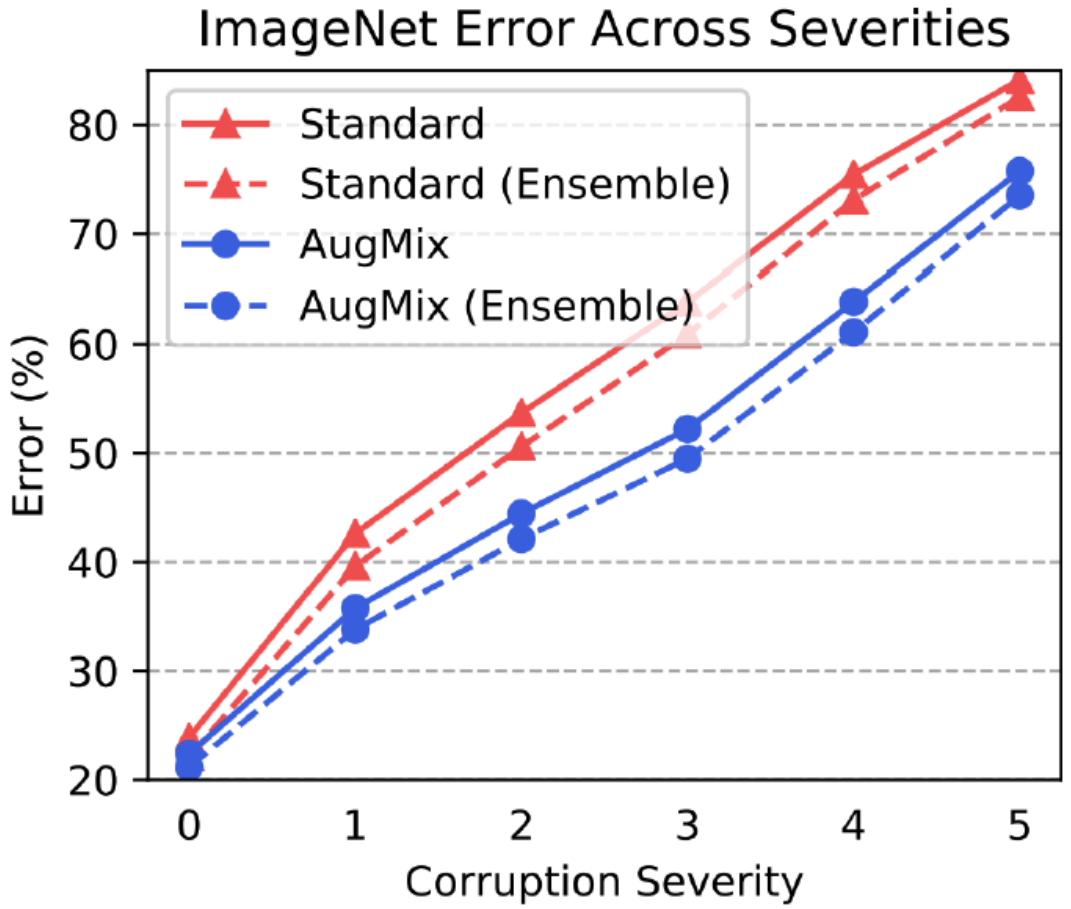


Figure 4: A realization of AUGMIX. Augmentation operations such as translate_x and weights such as m are randomly sampled. Randomly sampled operations and their compositions allow us to explore the semantically equivalent input space around an image. Mixing these images together produces a new image without veering too far from the original.

Data Augmentation



Hendrycks et al., AugMix: A Simple Data Processing Method to Improve Robustness and Uncertainty, 2020

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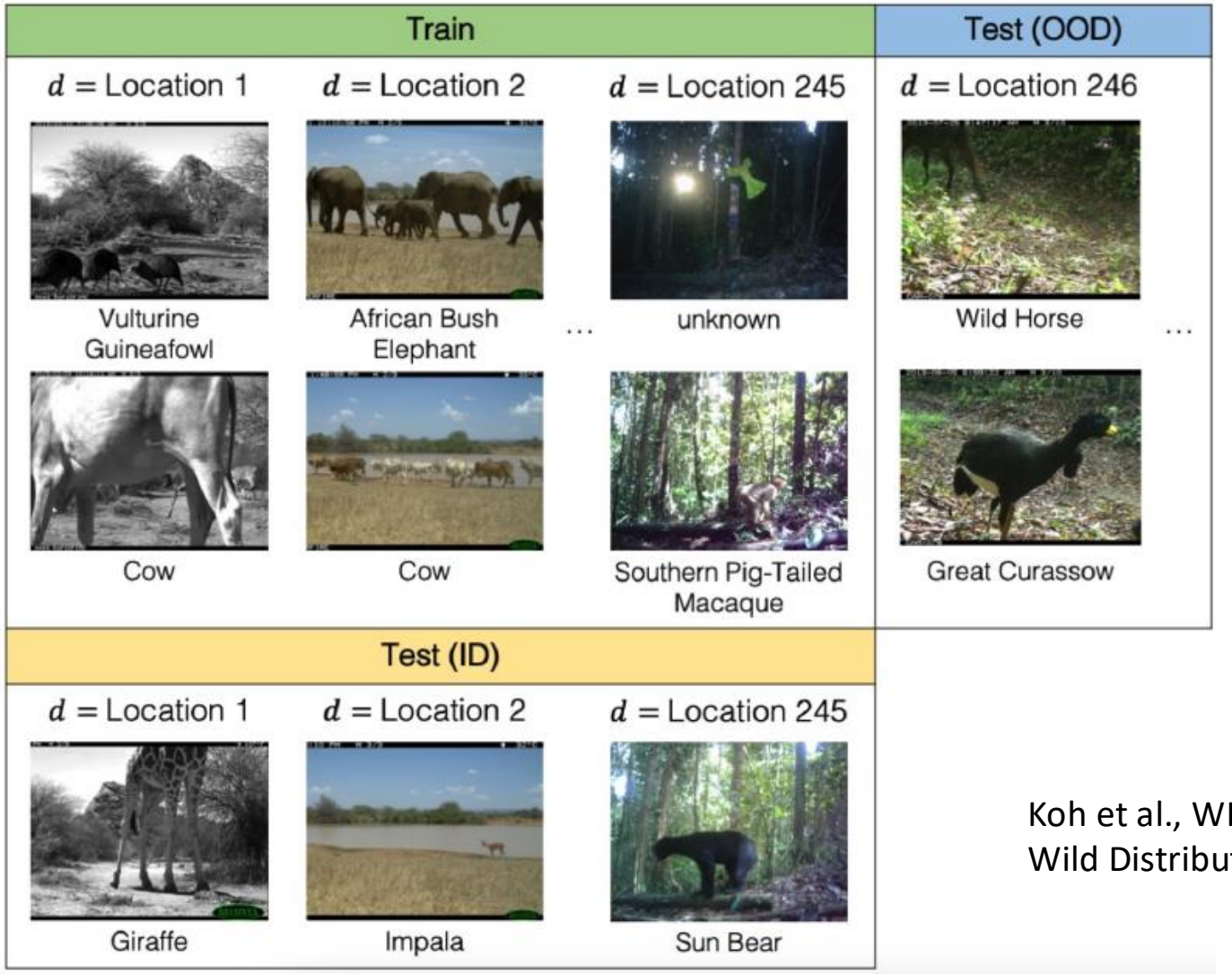
Example: Natural Language Processing

Article: Super Bowl 50
Paragraph: "*Peyton Manning became the first quarterback ever to lead two different teams to multiple Super Bowls. He is also the oldest quarterback ever to play in a Super Bowl at age 39. The past record was held by John Elway, who led the Broncos to victory in Super Bowl XXXIII at age 38 and is currently Denver's Executive Vice President of Football Operations and General Manager. Quarterback Jeff Dean had jersey number 37 in Champ Bowl XXXIV.*"
Question: "*What is the name of the quarterback who was 38 in Super Bowl XXXIII?*"
Original Prediction: John Elway
Prediction under adversary: Jeff Dean

Jia & Liang, Adversarial Examples for Evaluating Reading Comprehension Systems, 2021

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Example: Real Perturbations



Koh et al., WILDS: A Benchmark of in-the-Wild Distribution Shifts, 2020

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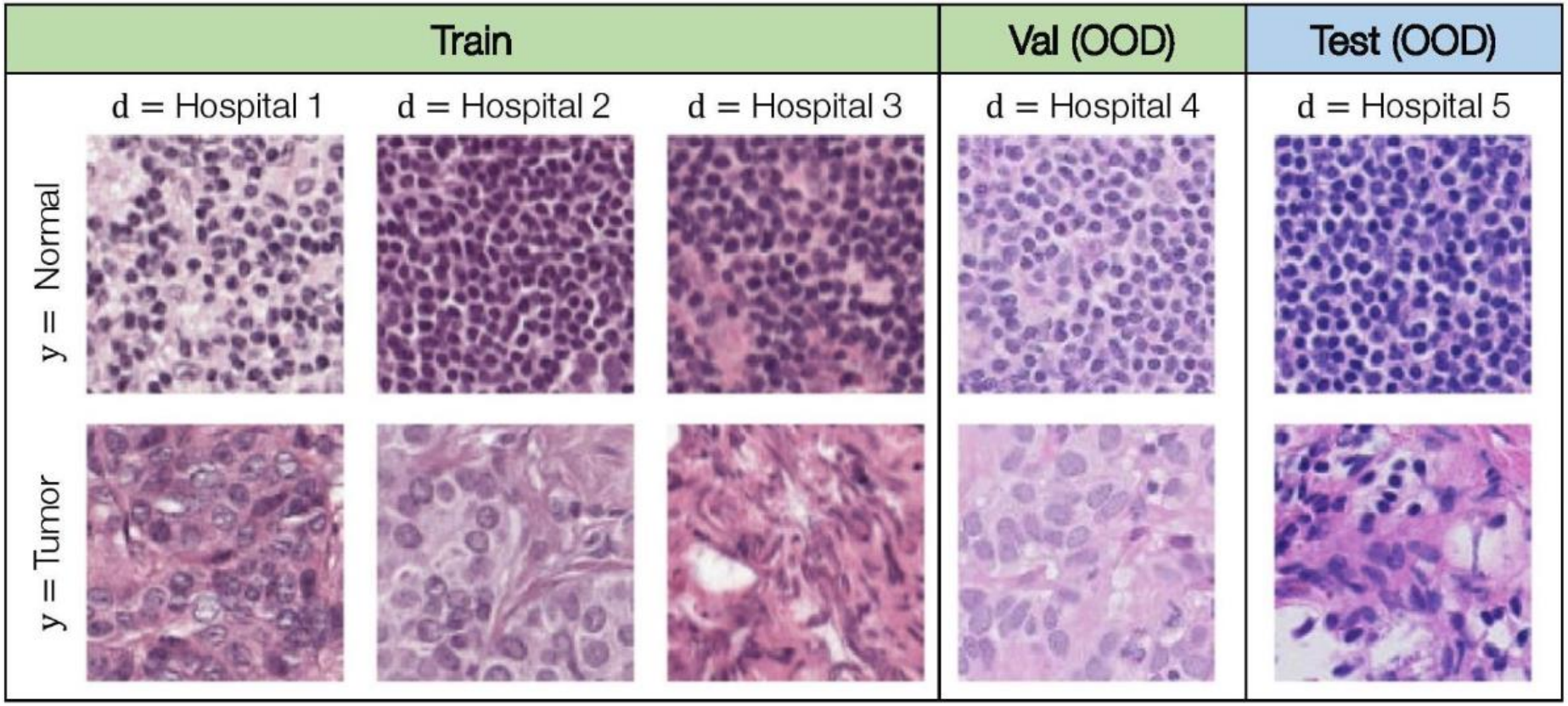
Example: Real Perturbations

	Train			Test	
Satellite Image (x)					
Year / Region (d)	2002 / Americas	2009 / Africa	2012 / Europe	2016 / Americas	2017 / Africa
Building / Land Type (y)	shopping mall	multi-unit residential	road bridge	recreational facility	educational institution

Koh et al., WILDS: A Benchmark of in-the-Wild Distribution Shifts, 2020

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Example: Real Perturbations



Koh et al., WILDS: A Benchmark of in-the-Wild Distribution Shifts, 2020

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Distribution Shift:
Notation & Problem Formulation

Traditional Supervised Learning

- **Problem setup**
  - Consider a parametric model family $\{f_\theta : \mathcal{X} \to \mathcal{Y} \mid \theta \in \Theta\}$
  - Consider a loss function $\ell(\theta; x, y)$
  - Consider a data distribution P over $\mathcal{X} \times \mathcal{Y}$

- **Supervised learning problem**
  - Given training dataset $Z \subseteq \mathcal{X} \times \mathcal{Y}$ consisting of i.i.d. samples $(x, y) \sim P$
  - Expected/empirical loss $\mathbb{E}_P[\ell(\theta; x, y)] \approx |Z|^{-1} \sum_{(x,y)\in Z} \ell(\theta; x, y)$
  - Goal is to compute $\hat{\theta} = \min_\theta |Z|^{-1} \sum_{(x,y)\in Z} \ell(\theta; x, y)$

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Distribution Shift

- **Distribution shift:** Training and test distributions differ
  - Training set consists of samples $(x_1, y_1), \ldots, (x_n, y_n) \sim P$
  - Test set consists of samples $(x'_1, y'_1), \ldots, (x'_m, y'_m) \sim Q$

- **Supervised learning under distribution shift**
  - Given training dataset $Z \subseteq \mathcal{X} \times \mathcal{Y}$ consisting of i.i.d. samples $(x, y) \sim P$
  - Goal is to minimize loss $\mathbb{E}_Q[\ell(\theta; x, y)] \not\approx |Z|^{-1} \sum_{(x,y)\in Z} \ell(\theta; x, y)$
  - Computing $\hat{\theta} = \min_\theta |Z|^{-1} \sum_{(x,y)\in Z} \ell(\theta; x, y)$ may not work

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Distribution Shift

- Intuitively, when can we hope to perform well on Q?

- **Impossible in general (what if we swap the labels?)**

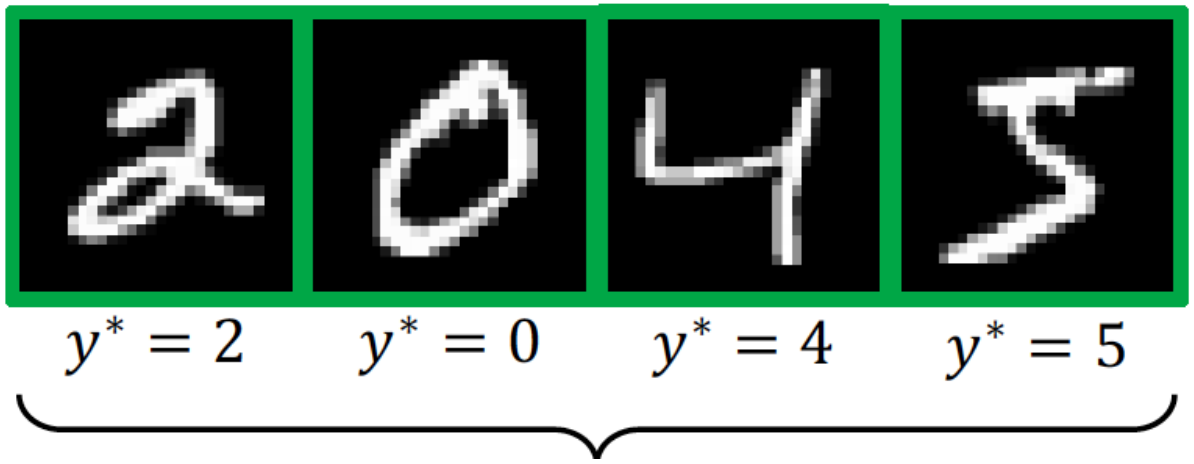
- Can we leverage additional information about the shift?
  - Make additional assumptions about shift
  - Leverage additional data
  - Both

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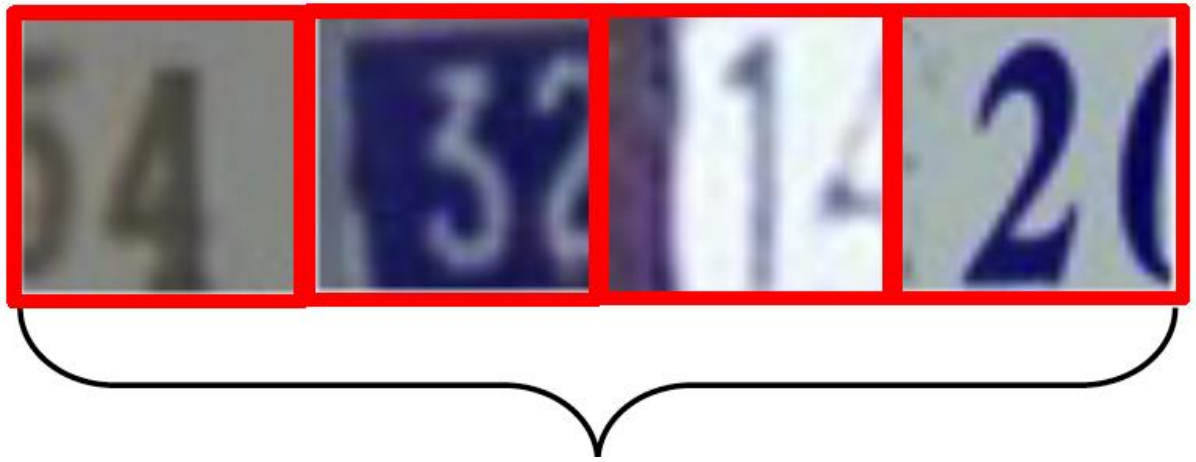
Unsupervised Domain Adaptation

- **Idea:** Use **some** information about the distribution shift

- Consider **unsupervised domain adaptation** setting



labeled training examples



unlabeled test examples
(possibly shifted)

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Unsupervised Domain Adaptation

- Data is easy to collect but labeling costs money
  - **Example:** Data from a different hospital

- Collect data during run time
  - **Example:** Self-driving car

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Types of Distribution Shift: Label Shift

$p(y) \neq q(y)$ but $p(x \mid y) = q(x \mid y)$

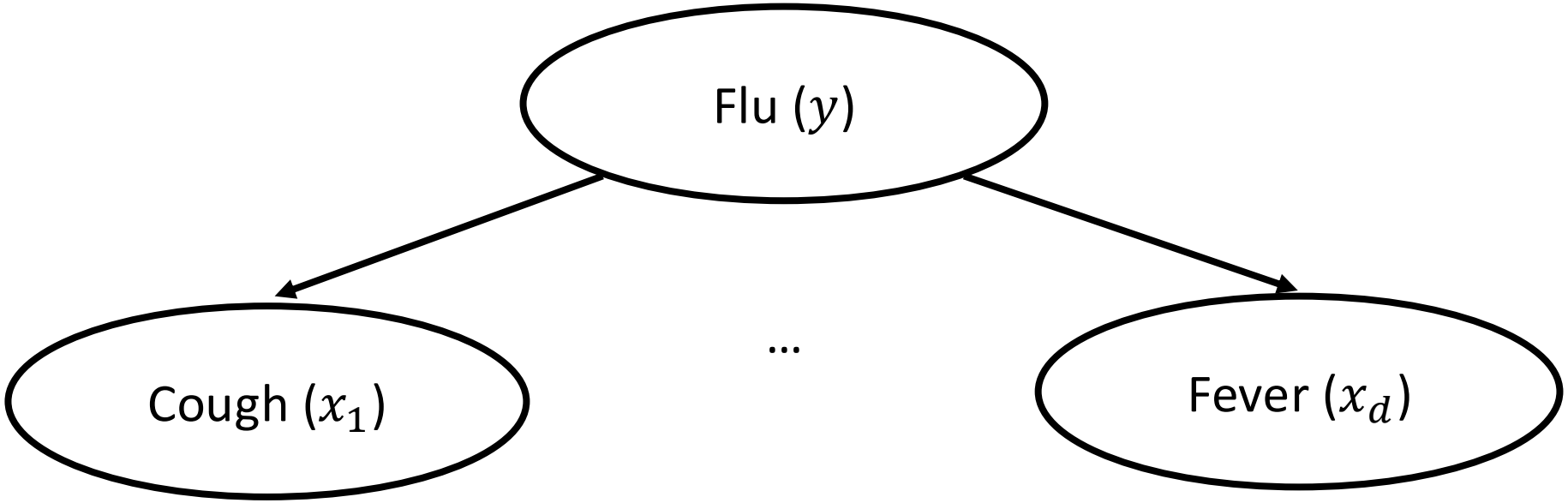
Label Shift Assumption

- Let p and q be the density functions for P and Q, respectively

- **Label Shift Assumption:** $p(y) \neq q(y)$
  - But $p(x \mid y) = q(x \mid y)$
  - **Intuition:** The rates of labels changes, but the kinds of labels don't

Label Shift Assumption

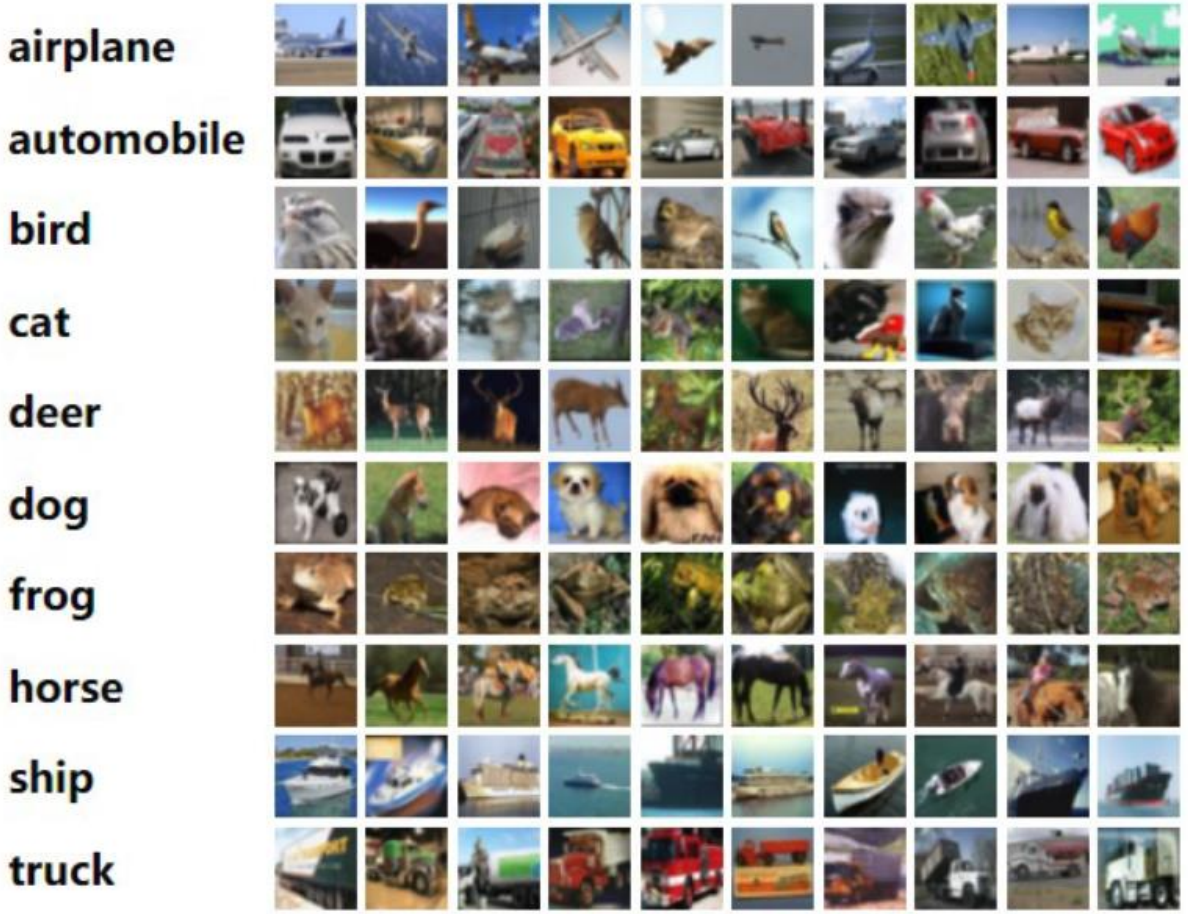
- **Example:** Increase in flu cases due to an outbreak
  - x are the symptoms, y is indicator for flu
  - $p(x \mid y)$ is rate of symptoms conditioned on having disease (stays the same)
  - $p(y)$ is rate of flu (can change if there is an outbreak)



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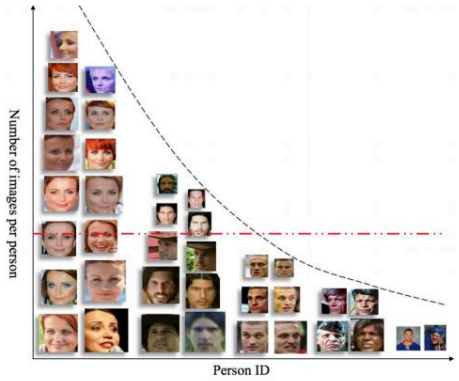
Label Shift Assumption

- **Example:** Changes in label distribution
  - x is an image, y is the label
  - $p(x \mid y)$ is the distribution of images of a given label
  - $p(y)$ is rate of that label
- Often, the training labels are balanced, which is a source of label shift

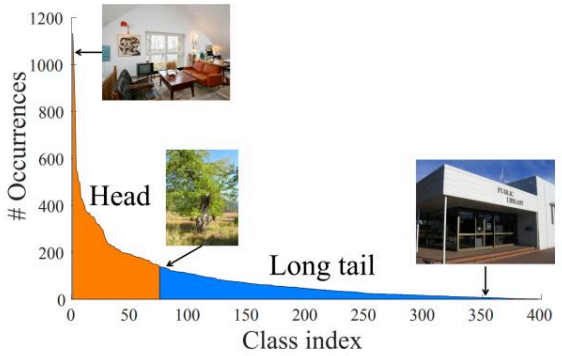


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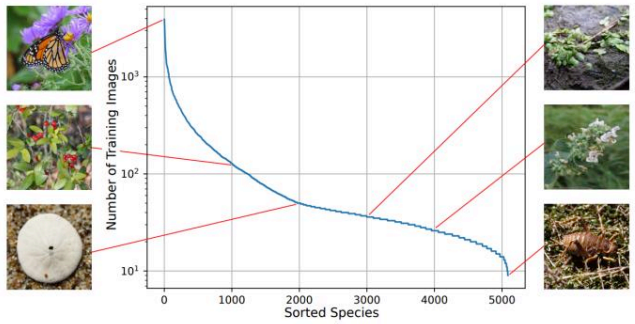
Class Imbalance



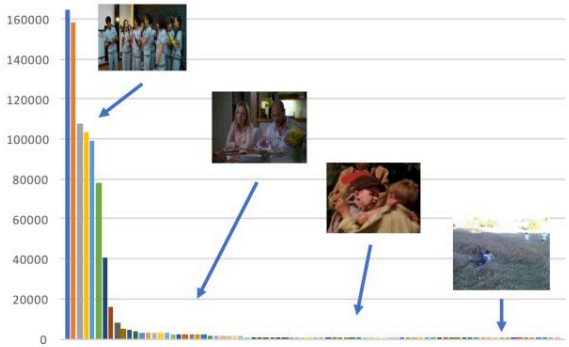
Faces [Zhang et al. 2017]



Places [Wang et al. 2017]

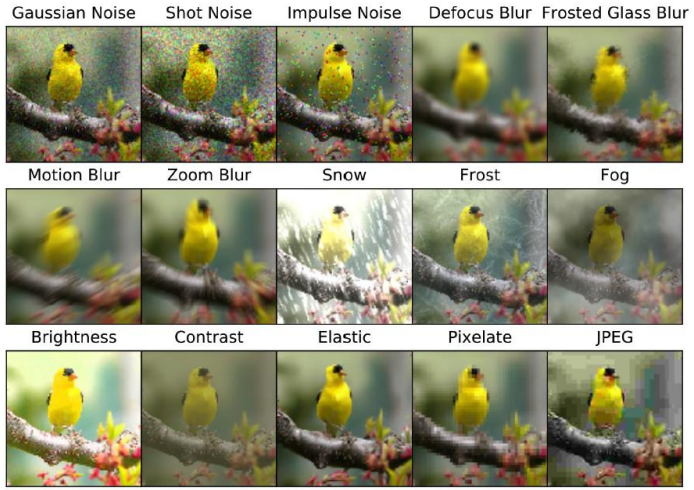


Species [Van Horn et al. 2019]



Actions [Zhang et al. 2019]

Source: https://liuziwei7.github.io/papers/longtail_slides.pdf



Hendrycks & Dietterich, Benchmarking Neural Network Robustness to Common Corruptions and Perturbations

Types of Distribution Shift: Covariate Shift

$$p(x) \neq q(x) \text{ but } p(y \mid x) = q(y \mid x)$$

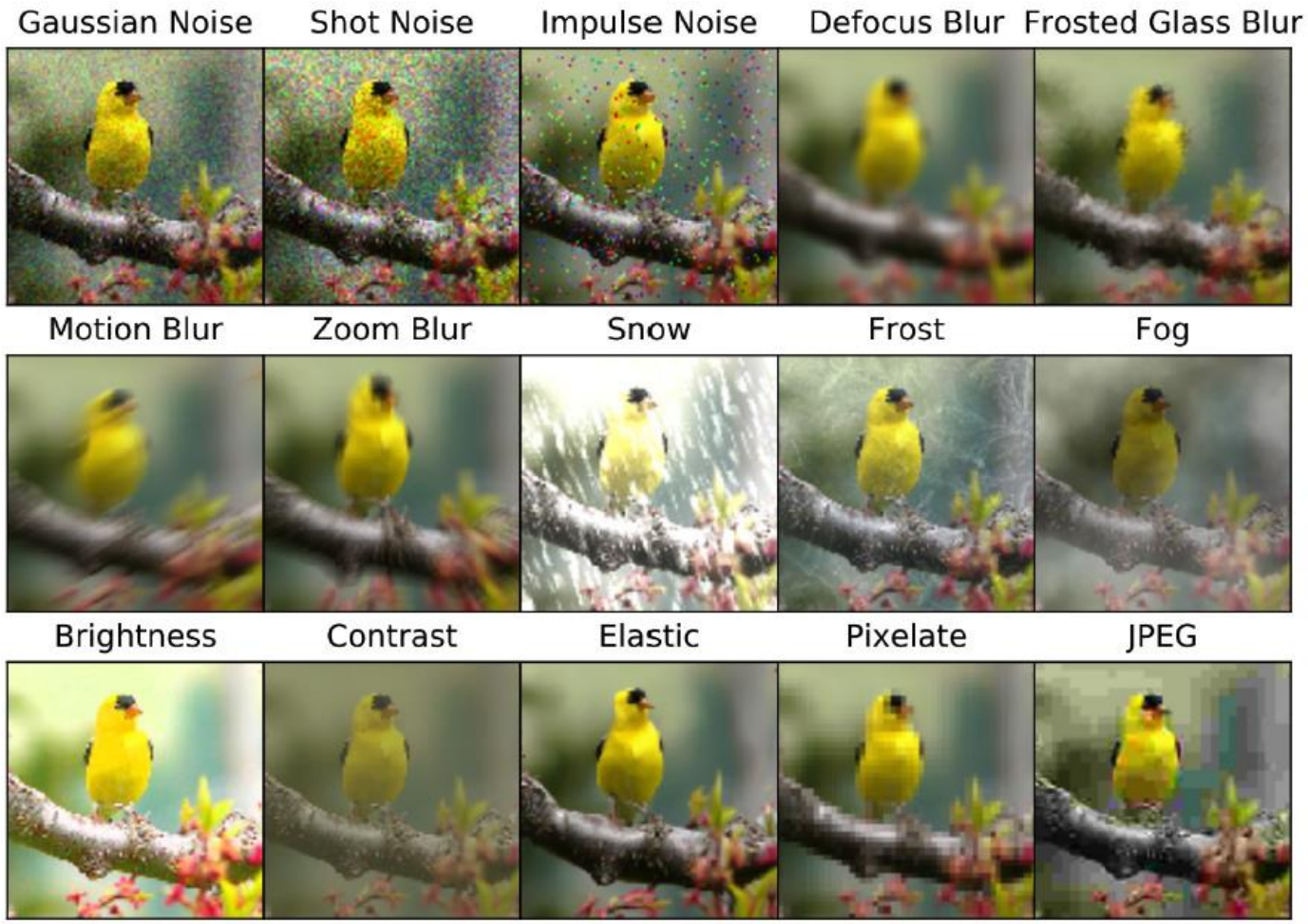
Covariate Shift Assumption

- Let p and q be the density functions for P and Q, respectively

- **Covariate Shift Assumption:** $p(x) \neq q(x)$
  - But $p(y \mid x) = q(y \mid x)$
  - **Intuition:** The label computation does not change, but the inputs can change

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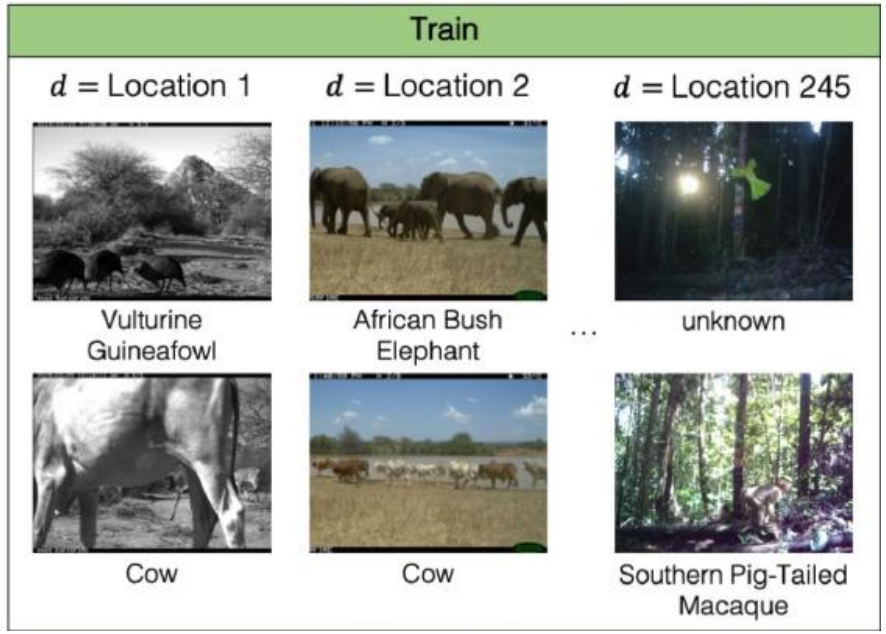
Covariate Shift Assumption



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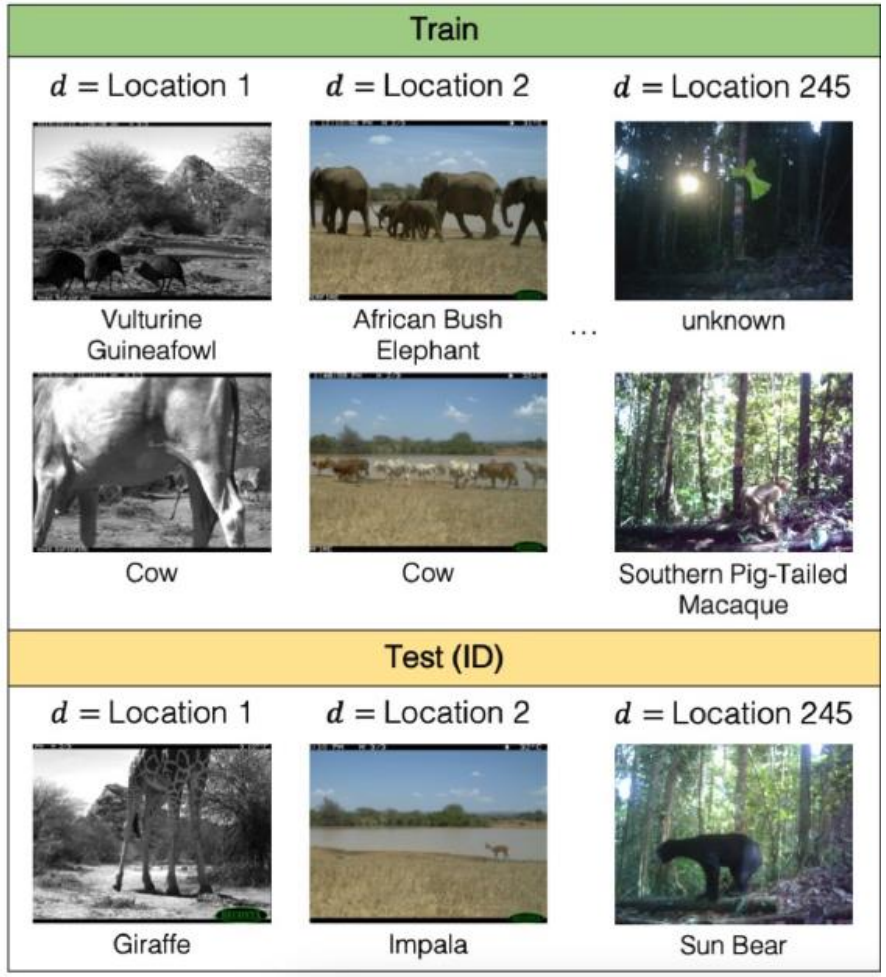
Covariate Shift Assumption



Koh et al., WILDS: A Benchmark of in-the-Wild Distribution Shifts, 2020

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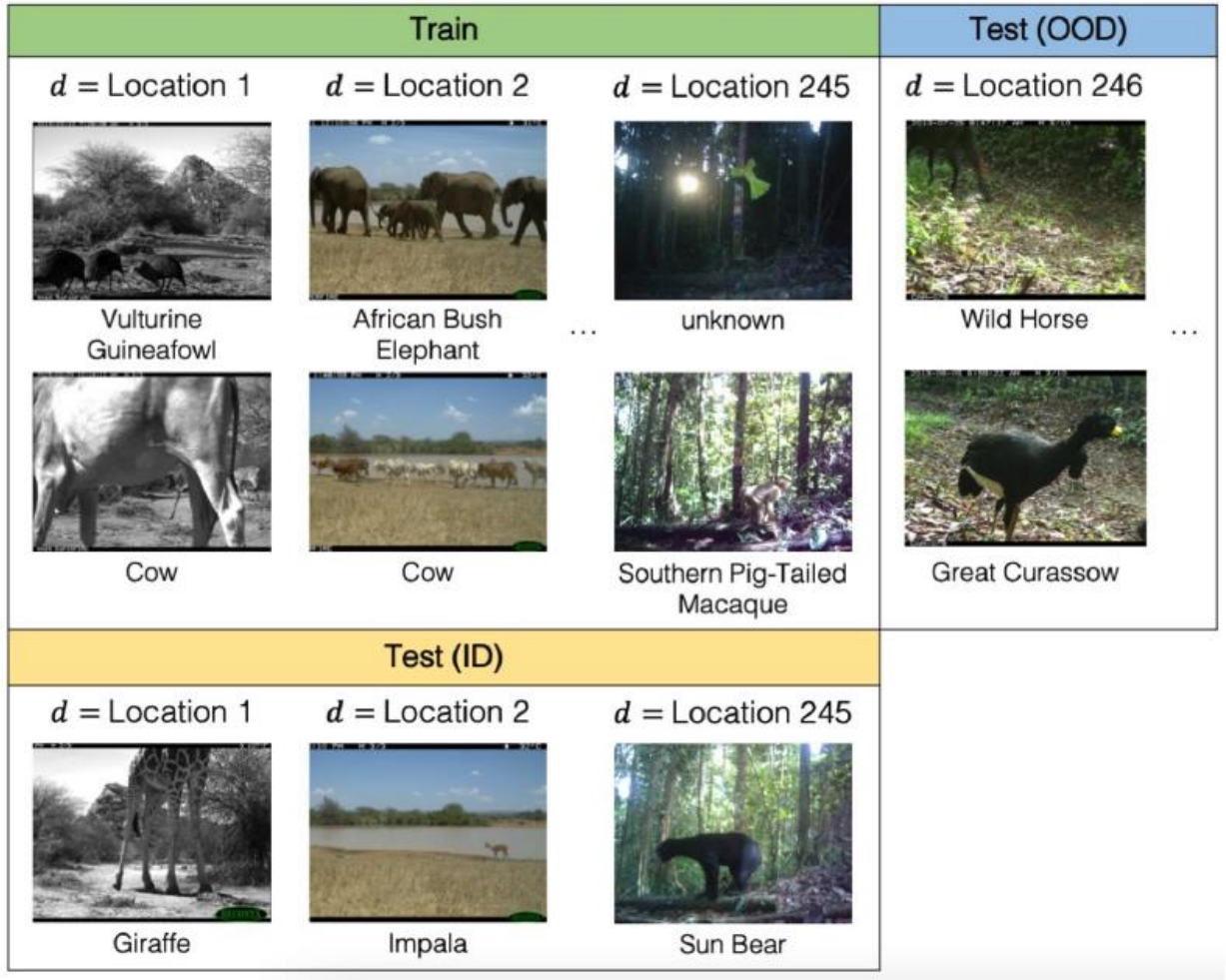
Covariate Shift Assumption



Koh et al., WILDS: A Benchmark of in-the-Wild Distribution Shifts, 2020

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Covariate Shift Assumption



Koh et al., WILDS: A Benchmark of in-the-Wild Distribution Shifts, 2020

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Covariate Shift Assumption

- **Computer vision**
  - Daytime vs. nighttime
  - Color shifts, lighting shifts, etc.
  - Driving in a new city

- **Natural language processing**
  - Change in vocabulary frequency over time
  - Regional vocabulary
  - News writing vs. conversational writing

- Covariate shift is pervasive

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Methods for Distributional Robustness

Importance Weighting

Importance Weighting

- Given distributions P and Q, the **importance weight (function)** is

$$w(x, y) = \frac{q(x, y)}{p(x, y)}$$

- **Key property (by definition):**

$$\mathbb{E}_Q[\ell(\theta; x, y)] = \mathbb{E}_P[\ell(\theta; x, y) \cdot w(x, y)]$$

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Importance Weighting

- Note that

$$\mathbb{E}_Q[\ell(\theta; x, y)] = \int_{X \times Y} \ell(\theta; x, y) \cdot q(x, y) \cdot dx \cdot dy$$

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Importance Weighting

- Note that

$$\begin{aligned}
\mathbb{E}_Q[\ell(\theta; x, y)] &= \int_{\mathcal{X}\times\mathcal{Y}} \ell(\theta; x, y) \cdot q(x, y) \cdot dx \cdot dy \\
&= \int_{\mathcal{X}\times\mathcal{Y}} \ell(\theta; x, y) \cdot \frac{q(x,y)}{p(x,y)} \cdot p(x, y) \cdot dx \cdot dy \\
&= \int_{\mathcal{X}\times\mathcal{Y}} \ell(\theta; x, y) \cdot w(x, y) \cdot p(x, y) \cdot dx \cdot dy \\
&= \mathbb{E}_P[\ell(\theta; x, y) \cdot w(x, y)]
\end{aligned}$$

- We have assumed the support of Q is contained in the support of P!

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Importance Weighting

- Given distributions P and Q, the **importance weight (function)** is

$$w(x, y) = \frac{q(x, y)}{p(x, y)}$$

- **Key property (by definition):**

$$\mathbb{E}_Q[\ell(\theta; x, y)] = \mathbb{E}_P[\ell(\theta; x, y) \cdot w(x, y)]$$

- **Key question:** How to compute importance weights?

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Importance Weights for Label Shift

- In the label shift setting, we have

$$
\begin{aligned}
w(x,y) &= \frac{q(x,y)}{p(x,y)} \\
&= \frac{q(x \mid y)q(y)}{p(x \mid y)p(y)} \\
&= \frac{q(y)}{p(y)} \\
&:= w(y)
\end{aligned}
$$

Label Shift Assumption

$p(y) \neq q(y)$ but $p(x \mid y) = q(x \mid y)$

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Importance Weights for Label Shift

- If we know $w(y)$, then we have

$$\mathbb{E}_Q[\ell(\theta; x, y)] = \mathbb{E}_P[\ell(\theta; x, y) \cdot w(x, y)] = \mathbb{E}_P[\ell(\theta; x, y) \cdot w(y)]$$

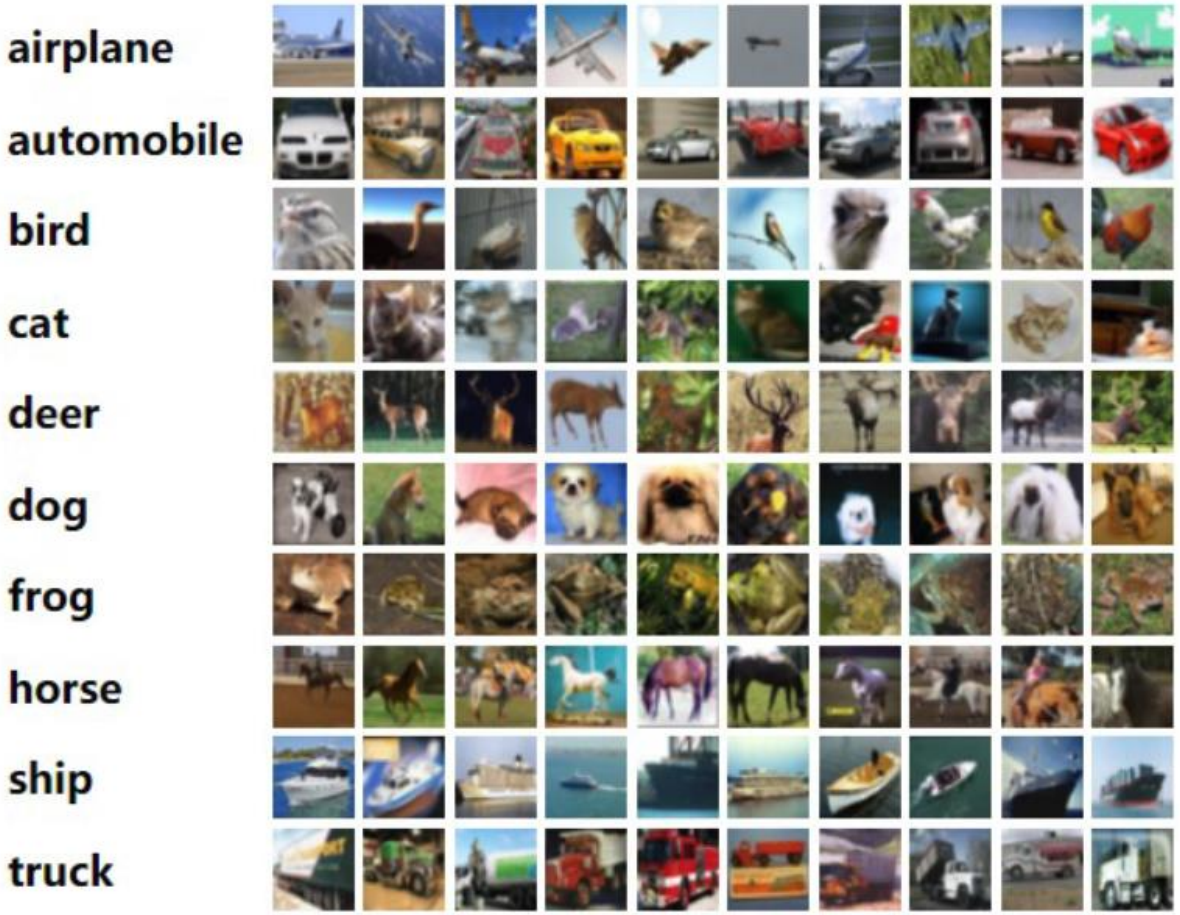
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Label Shift Assumption

$$\mathbb{E}_Q[\ell(\theta; x, y)] = \mathbb{E}_P[\ell(\theta; x, y) \cdot w(x, y)] = \mathbb{E}_P[\ell(\theta; x, y) \cdot w(y)]$$

- **Training:** $p(y) = \frac{1}{10}$
- **Test:** $q(\text{automobile}) = \frac{1}{2}$ and $q(y) = \frac{1}{18}$ otherwise
- Then, the loss might be:

$$\frac{10}{18} \cdot \ell(x_1, \text{dog}) + \frac{10}{2} \cdot \ell(x_2, \text{automobile}) + \frac{10}{18} \cdot \ell(x_3, \text{frog})$$



Intuition: Automobile class was underrepresented in the source distribution. We give higher weight to its loss calculation to compensate.

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Importance Weights for Label Shift

- If we know $w(y)$, then we have

$$\mathbb{E}_Q[\ell(\theta; x, y)] = \mathbb{E}_P[\ell(\theta; x, y) \cdot w(x, y)] = \mathbb{E}_P[\ell(\theta; x, y) \cdot w(y)]$$

- How do we compute $w(y)$?



Challenge: Challenge: We do not know the correct labels during inference (Q)

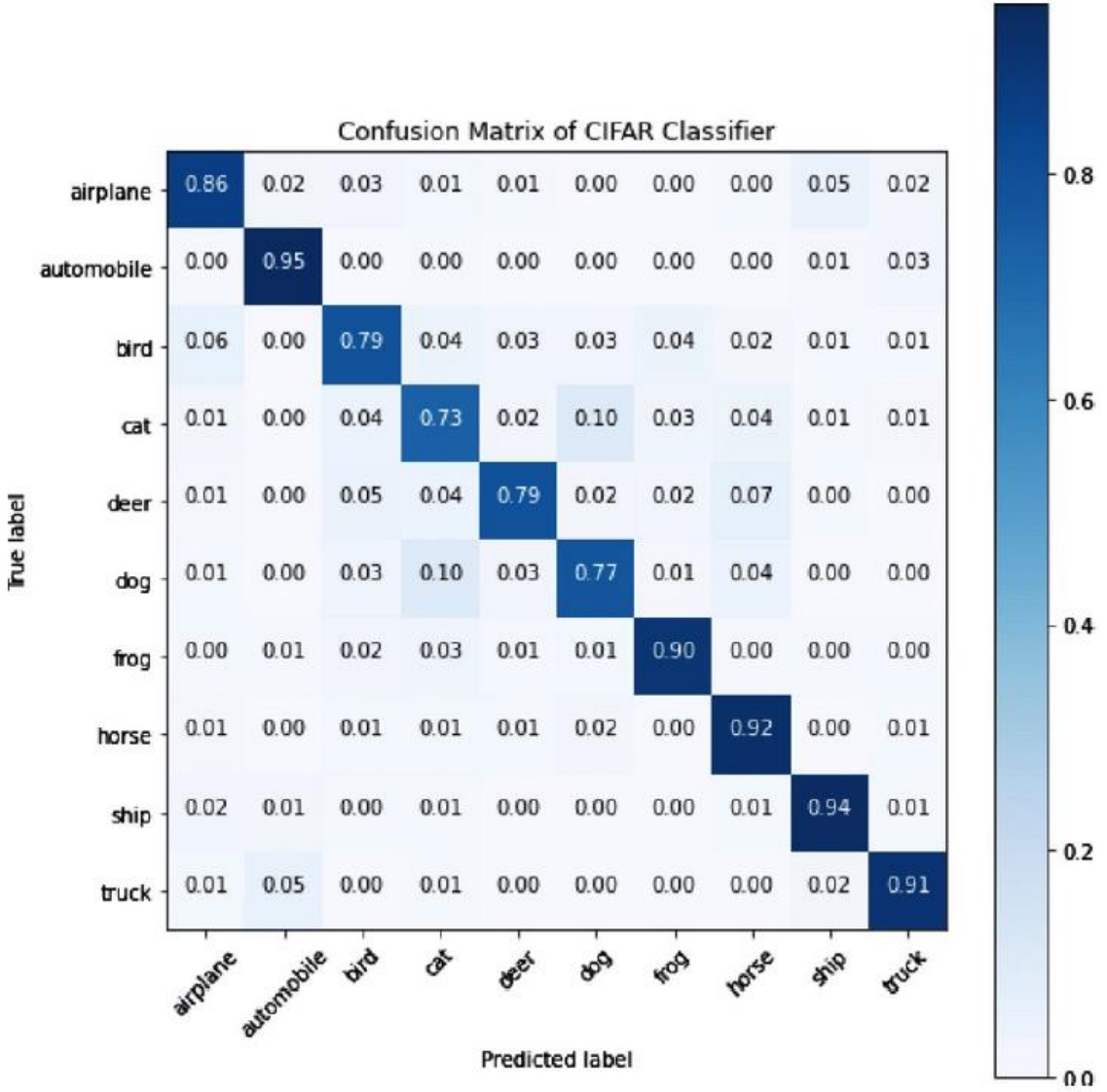
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Importance Weights for Label Shift

- Given a classifier $f: \mathcal{X} \to \mathcal{Y}$, where $\mathcal{Y} = \{1, \ldots, K\}$, consider the confusion matrix $C \in \mathbb{R}^{K \times K}$ defined by

$$C_{ij} = \mathbb{P}_P[f(x) = i, y = j]$$

- Also, define $p, q \in \mathbb{R}^K$ as prediction probabilities by:
  - $p_i = \mathbb{P}_P[f(x) = i]$
  - $q_i = \mathbb{P}_Q[f(x) = i]$



Sooksatra, Evaluation of adversarial attacks sensitivity of classifiers with occluded input data

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Importance Weights for Label Shift

$$C_{ij} = \mathbb{P}_P[f(x) = i, y = j] \quad p_i = \mathbb{P}_P[f(x) = i] \quad q_i = \mathbb{P}_Q[f(x) = i]$$

- Since $f(x)$ only depends on x, we have

$$\mathbb{P}_P[f(x) = i \mid y = j] = \int_X 1(f(x) = i) \cdot p(x \mid y = j) \cdot dx$$

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Importance Weights for Label Shift

$$C_{ij} = \mathbb{P}_P[f(x) = i, y = j] \quad p_i = \mathbb{P}_P[f(x) = i] \quad q_i = \mathbb{P}_Q[f(x) = i]$$

- Since $f(x)$ only depends on x, we have

$$\begin{aligned}\mathbb{P}_P[f(x) = i \mid y = j] &= \int_X 1(f(x) = i) \cdot p(x \mid y = j) \cdot dx \\ &= \int_X 1(f(x) = i) \cdot q(x \mid y = j) \cdot dx \\ &= \mathbb{P}_Q[f(x) = i \mid y = j]\end{aligned}$$



In other words, the conditional confusion matrix stays the same for a fixed f(x), i.e., a decision boundary.

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Importance Weights for Label Shift

$$C_{ij} = \mathbb{P}_P[f(x) = i, y = j] \quad p_i = \mathbb{P}_P[f(x) = i] \quad q_i = \mathbb{P}_Q[f(x) = i]$$

$$\mathbb{P}_P[f(x) = i \mid y = j] = \mathbb{P}_Q[f(x) = i \mid y = j]$$

- Now, we have

$$q_i = \mathbb{P}_Q[f(x) = i] = \sum_{j=1}^{k} \mathbb{P}_Q[f(x) = i \mid y = j] \cdot \mathbb{P}_Q[y = j]$$

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Importance Weights for Label Shift

$$C_{ij} = \mathbb{P}_P[f(x) = i, y = j] \quad p_i = \mathbb{P}_P[f(x) = i] \quad q_i = \mathbb{P}_Q[f(x) = i]$$

$$\mathbb{P}_P[f(x) = i \mid y = j] = \mathbb{P}_Q[f(x) = i \mid y = j]$$

- Now, we have

$$\begin{aligned} q_i &= \sum_{j=1}^{k} \mathbb{P}_Q[f(x) = i \mid y = j] \cdot \mathbb{P}_Q[y = j] \\ &= \sum_{j=1}^{k} \mathbb{P}_P[f(x) = i \mid y = j] \cdot \mathbb{P}_Q[y = j] \\ &= \sum_{j=1}^{k} \frac{\mathbb{P}_P[f(x)=i, y=j]}{\mathbb{P}_P[y=j]} \cdot \mathbb{P}_Q[y = j] \\ &= \sum_{j=1}^{k} \mathbb{P}_P[f(x) = i, y = j] \cdot \frac{\mathbb{P}_Q[y=j]}{\mathbb{P}_P[y=j]} \end{aligned}$$

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Importance Weights for Label Shift

$$C_{ij} = \mathbb{P}_P[f(x) = i, y = j] \quad p_i = \mathbb{P}_P[f(x) = i] \quad q_i = \mathbb{P}_Q[f(x) = i]$$

$$\mathbb{P}_P[f(x) = i \mid y = j] = \mathbb{P}_Q[f(x) = i \mid y = j]$$

- Now, we have

$$q_i = \sum_{j=1}^{k} \mathbb{P}_P[f(x) = i, y = j] \cdot \boxed{\frac{\mathbb{P}_Q[y=j]}{\mathbb{P}_P[y=j]}} \quad w(j)$$

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Importance Weights for Label Shift

$$C_{ij} = \mathbb{P}_P[f(x) = i, y = j] \quad p_i = \mathbb{P}_P[f(x) = i] \quad q_i = \mathbb{P}_Q[f(x) = i]$$

$$\mathbb{P}_P[f(x) = i \mid y = j] = \mathbb{P}_Q[f(x) = i \mid y = j]$$

- Now, we have

$$q_i = \sum_{j=1}^{K} \mathbb{P}_P[f(x) = i, y = j] \cdot w(j)$$

- For all q_i, we can write this as matrix multiplication:

$$\begin{bmatrix} q_1 \\ \vdots \\ q_K \end{bmatrix} = \begin{bmatrix} \mathbb{P}_P[f(x) = 1, y = 1] & \cdots & \mathbb{P}_P[f(x) = 1, y = K] \\ \vdots & \ddots & \vdots \\ \mathbb{P}_P[f(x) = K, y = 1] & \cdots & \mathbb{P}_P[f(x) = K, y = K] \end{bmatrix} \begin{bmatrix} w(1) \\ \vdots \\ w(K) \end{bmatrix}$$

$$q = Cw \Rightarrow w = C^{-1}q$$

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Importance Weights for Label Shift

$$C_{ij} = \mathbb{P}_P[f(x) = i, y = j] \quad p_i = \mathbb{P}_P[f(x) = i] \quad q_i = \mathbb{P}_Q[f(x) = i]$$
$$\mathbb{P}_P[f(x) = i \mid y = j] = \mathbb{P}_Q[f(x) = i \mid y = j]$$

- Now, we have

$$w = C^{-1}q$$

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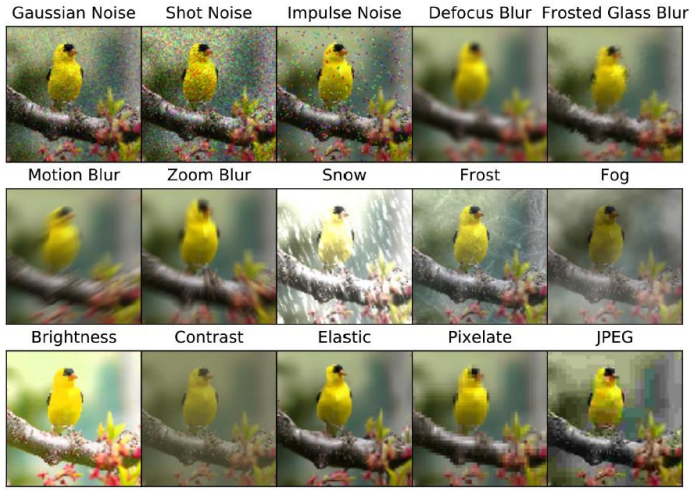
Supervised Learning with Label Shift

- **Input:** Training dataset Z, unlabeled test dataset X
- **Step 1:** Train f on Z
- **Step 2:** Estimate using the dataset:
  - $C_{ij} = \mathbb{P}_P[f(x) = i, y = j] \approx |Z|^{-1} \sum_{(x,y)\in Z} 1(f(x) = i \wedge y = j)$
  - $q_i = \mathbb{P}_Q[f(x) = i] \approx |X|^{-1} \sum_{x\in X} 1(f(x) = i)$
- **Step 3:** Compute $w = C^{-1}q$
- **Step 4:** Compute $\hat{\theta} = \arg\min_{\theta} \sum_{(x,y)\in Z} \ell(\theta; x, y) \cdot w(y)$

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Tutorial on Label Shift

https://colab.research.google.com/drive/1fpxfcIJW5UxxX72fsS98a0fkXSEeTRGr?usp=drive_link



Hendrycks & Dietterich, Benchmarking Neural Network Robustness to Common Corruptions and Perturbations

Importance Weights for Covariate Shift

$p(x) \neq q(x)$ but $p(y \mid x) = q(y \mid x)$

Importance Weights for Covariate Shift

- In the covariate shift setting, we have

$$w(x,y) = \frac{q(x,y)}{p(x,y)} = \frac{q(y|x)q(x)}{p(y|x)p(x)}$$

Remember:

- In the label shift setting, we have

$$w(x,y) = \frac{q(x,y)}{p(x,y)} = \frac{q(x|y)q(y)}{p(x|y)p(y)} = \frac{q(y)}{p(y)} := w(y)$$

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Importance Weights for Covariate Shift

- In the covariate shift setting, we have

$$
\begin{aligned}
w(x,y) &= \frac{q(x,y)}{p(x,y)} \\
&= \frac{q(y|x)q(x)}{p(y|x)p(x)} \\
&= \frac{q(x)}{p(x)} \\
&:= w(x)
\end{aligned}
$$

Remember:

- In the label shift setting, we have

$$
\begin{aligned}
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&= \frac{q(y)}{p(y)} \\
&:= w(y)
\end{aligned}
$$

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Importance Weights for Covariate Shift

- If we know $w(x)$, then we have

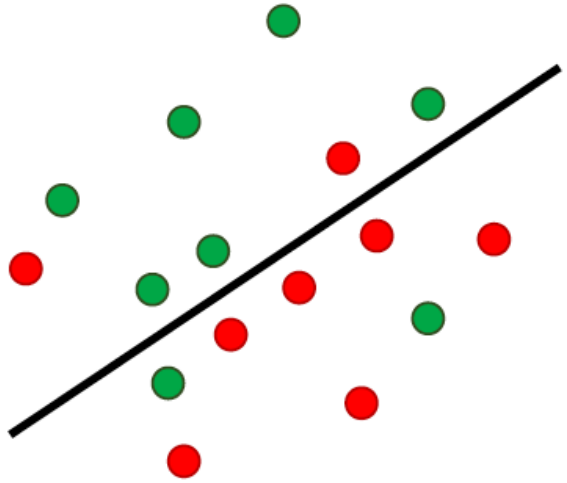
$$\mathbb{E}_Q[\ell(\theta; x, y)] = \mathbb{E}_P[\ell(\theta; x, y) \cdot w(x, y)] = \mathbb{E}_P[\ell(\theta; x, y) \cdot w(x)]$$

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Importance Weights for Covariate Shift

Estimating the weights

- Define a new distribution R over $\{0,1\}\times\mathcal{X}$:
  - Sample $b \sim \text{Bernoulli}\left(\frac{1}{2}\right)$
  - If $b = 0$, then sample $(x,\cdot) \sim P$
  - If $b = 1$, then sample $(x,\cdot) \sim Q$



- Suppose we know $r(b \mid x)$, then by Bayes' rule, we have

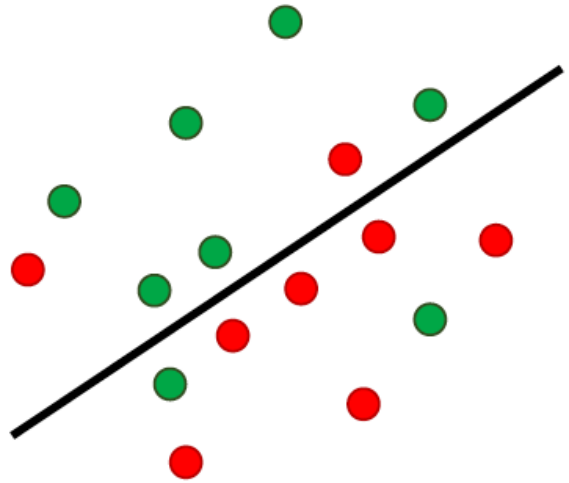
$$r(b = 0 \mid x) = \frac{r(x \mid b = 0)r(b = 0)}{r(x \mid b = 0)r(b = 0) + r(x \mid b = 1)r(b = 1)}$$

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Importance Weights for Covariate Shift

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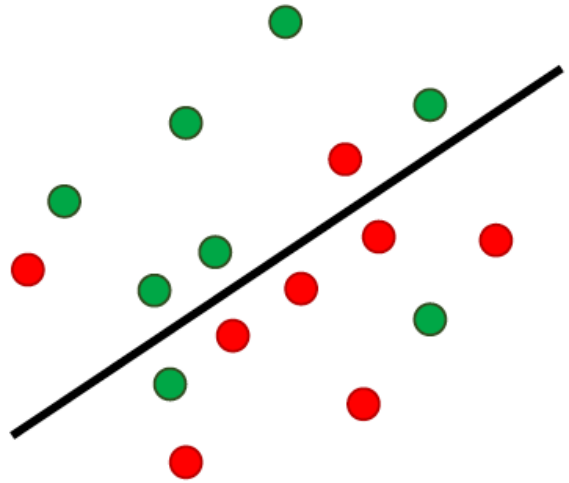
$$r(b = 0 \mid x) = \frac{r(x \mid b = 0) \cdot \frac{1}{2}}{r(x \mid b = 0) \cdot \frac{1}{2} + r(x \mid b = 1) \cdot \frac{1}{2}}$$

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Importance Weights for Covariate Shift

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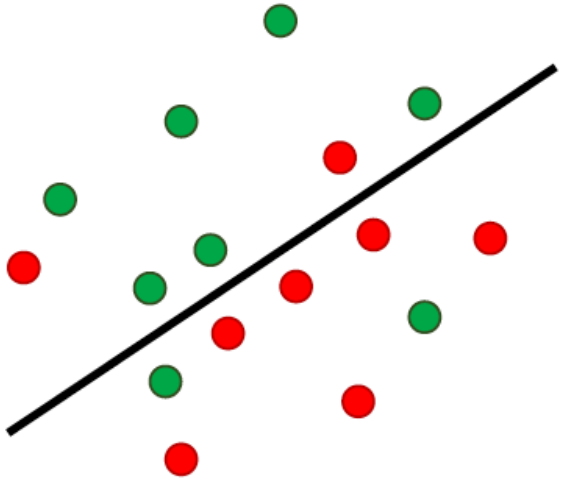
$$r(b = 0 \mid x) = \frac{r(x \mid b = 0)}{r(x \mid b = 0) + r(x \mid b = 1)}$$

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Importance Weights for Covariate Shift

Estimating the weights

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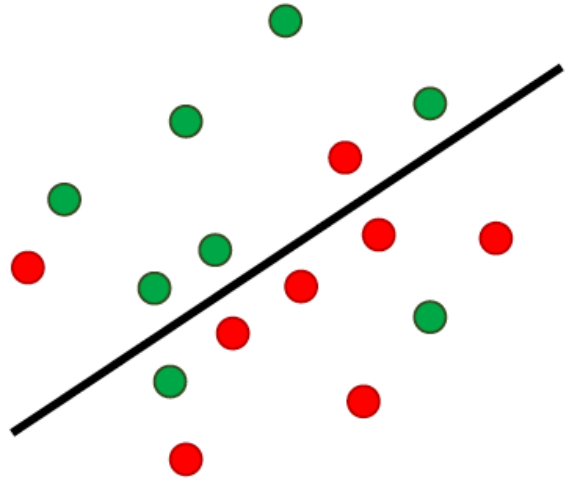
$$r(b = 0 \mid x) = \frac{p(x)}{p(x) + q(x)}$$

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Importance Weights for Covariate Shift

Estimating the weights

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  - Sample $b \sim \text{Bernoulli}\left(\frac{1}{2}\right)$
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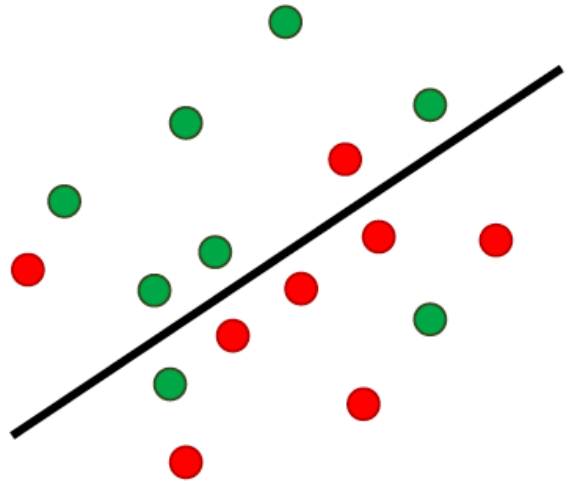
$$r(b = 0 \mid x) = \frac{1}{1 + \frac{q(x)}{p(x)}}$$

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Importance Weights for Covariate Shift

Estimating the weights

- Define a new distribution R over $\{0,1\}\times\mathcal{X}$:
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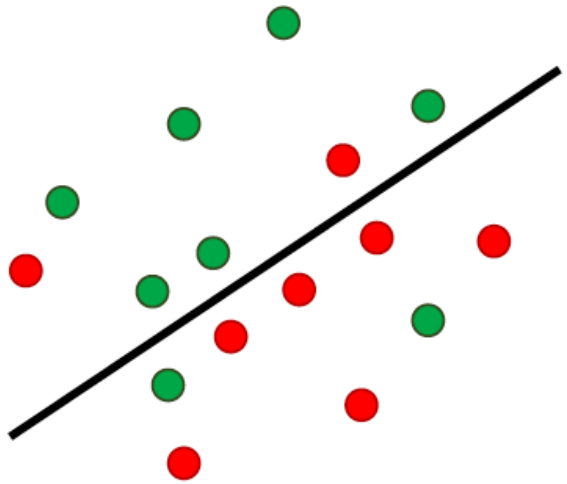
$$r(b = 0 \mid x) = \frac{1}{w(x) + 1}$$

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Importance Weights for Covariate Shift

Estimating the weights

- Define a new distribution R over $\{0,1\}\times\mathcal{X}$:
  - Sample $b \sim \text{Bernoulli}\left(\frac{1}{2}\right)$
  - If $b = 0$, then sample $(x,\cdot) \sim P$
  - If $b = 1$, then sample $(x,\cdot) \sim Q$



- Suppose we know $r(b \mid x)$, then by Bayes' rule, we have

$$w(x) + 1 = \frac{1}{r(b = 0 \mid x)}$$

$$\Rightarrow \quad w(x) = \frac{1}{r(b = 0 \mid x)} - 1$$

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Estimating Source-Target Probability

- We can construct a dataset of i.i.d. samples $(x, b) \sim R$
  - For simplicity, assume that $|X| = |Z|$
  - Then, consider

$$X' = \{(x, 0) \mid (x, y) \in Z\} \cup \{(x, 1) \mid x \in X\}$$

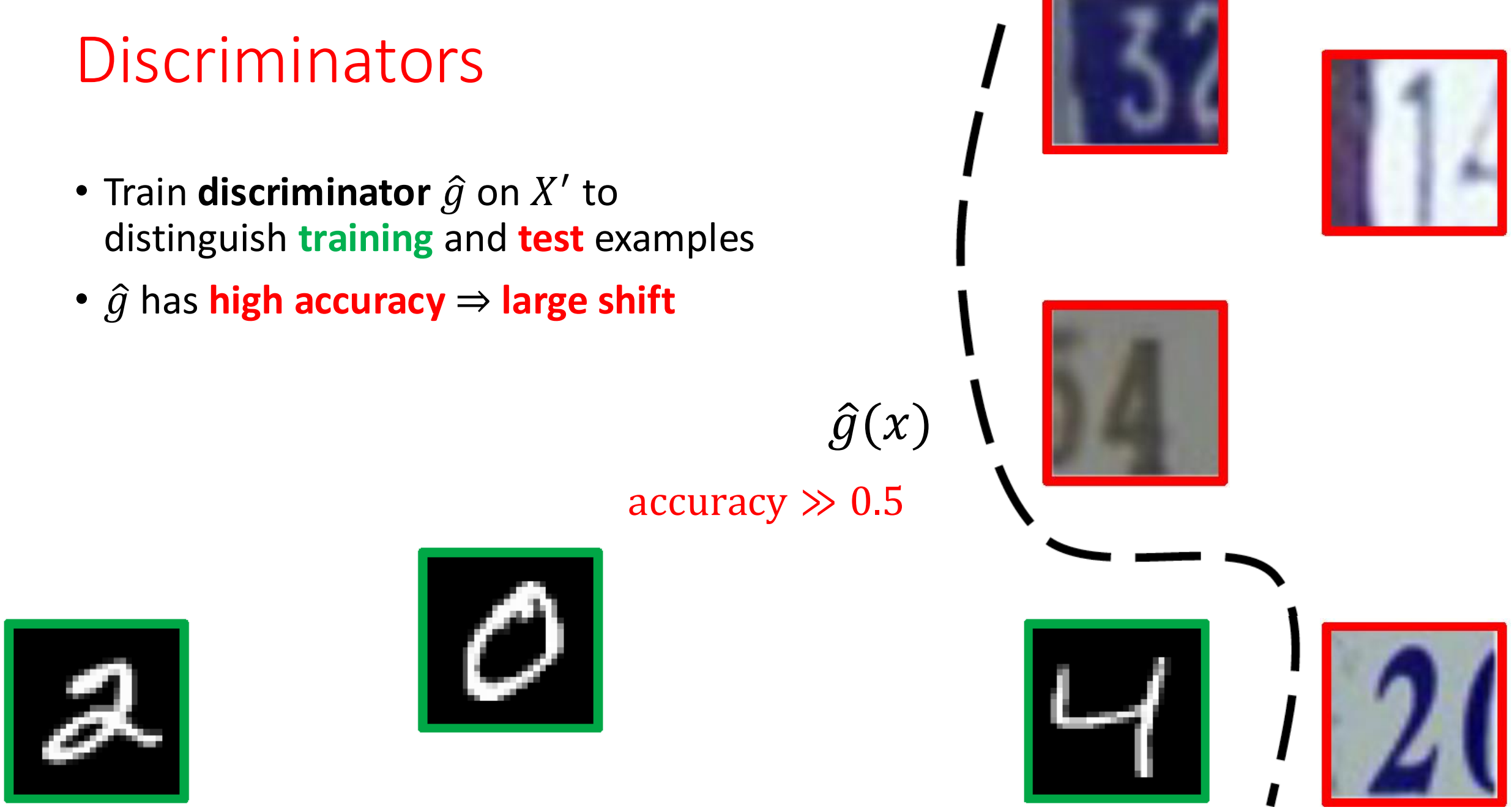
  - This dataset consists of i.i.d. samples $(x, b) \sim R$

- Given i.i.d. samples $(x, b) \sim R$, then $r(b = 1 \mid x)$ is the same as the probability of "label" b given "input" x
  - **Idea:** Train a model (called a **discriminator**) on X' to predict b given x

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Discriminators

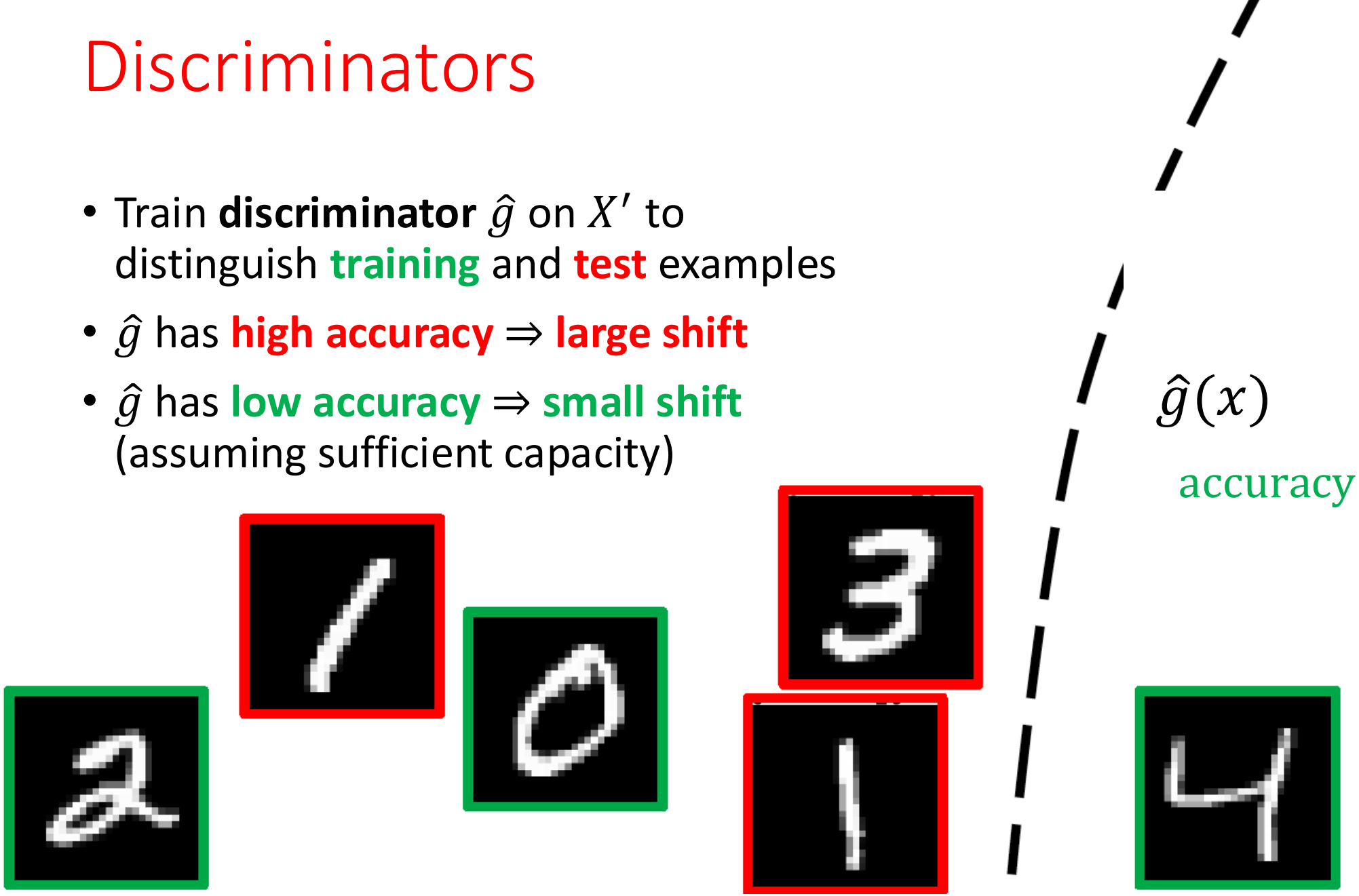
- Train **discriminator** $\hat{g}$ on X' to distinguish **training** and **test** examples
- $\hat{g}$ has **high accuracy** ⇒ **large shift**



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Discriminators

- Train **discriminator** $\hat{g}$ on X' to distinguish **training** and **test** examples
- $\hat{g}$ has **high accuracy** ⇒ **large shift**
- $\hat{g}$ has **low accuracy** ⇒ **small shift** (assuming sufficient capacity)



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Supervised Learning with Covariate Shift

- **Input:** Training dataset Z, unlabeled test dataset X
- **Step 1:** Construct $X' = \{(x, 0) \mid (x, y) \in Z\} \cup \{(x, 1) \mid x \in X\}$ and train $\hat{g}$ on X' to predict b given x
- **Step 2:** Compute $w(x) = \frac{1}{\hat{g}(b=0 \mid x)} - 1$
- **Step 3:** Compute $\hat{\theta} = \arg\min_{\theta} \sum_{(x,y)\in Z} \ell(\theta; x, y) \cdot w(x)$

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Importance Weights

- **Pros:**
  - Principled technique for addressing distribution shift
  - "Granular" quantification of shift (obtain an estimate of the shift for each example, not just just the overall shift)

- **Cons:**
  - Does not work when support of Q is not contained in support of P
  - Even if the above is satisfied, importance weights are large if $P(x, y)$ is small

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Alternatives to Importance Weighting

- Covariate shift
  - Domain-invariant representation learning, e.g., through disentanglement or adversarial training.
  - After training on source data => self-training with high-confidence unlabeled target data.
  - Distributionally Robust Optimization (DRO)

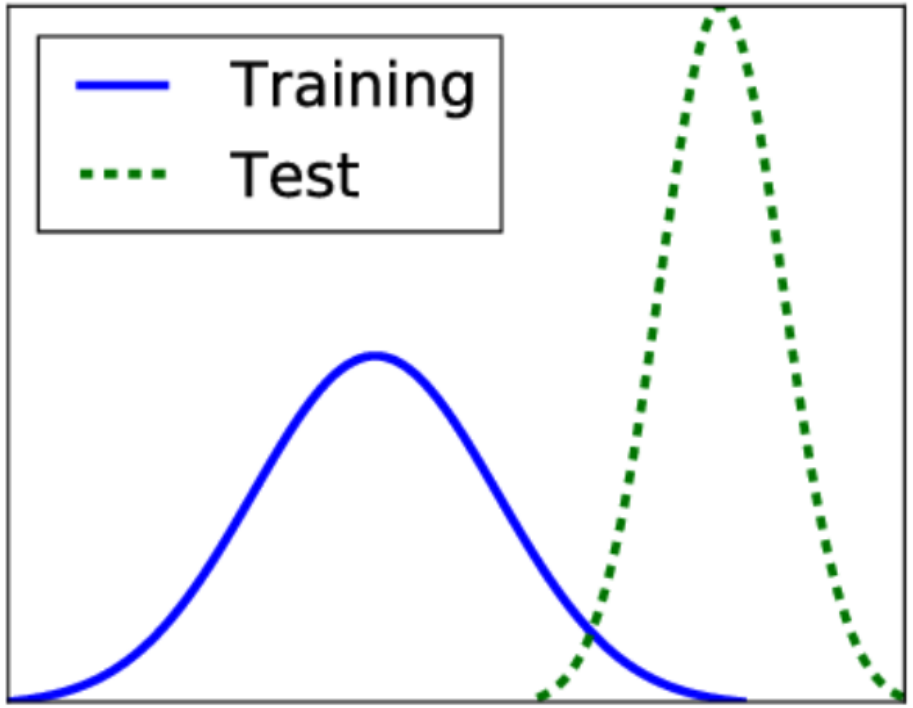
- Label shift
  - Expectation maximization: Estimate labels on Q (E-step) then modify class priors (M-step)
  - Group Distributionally Robust Optimization (Group DRO) by treating each class as a group and performing DRO for each class.

Evaluation Bounds for Distributional Robustness

and Distributionally Robust Optimization

Support of Shifted Data

- **Assumption:** Support of Q is not contained in support of P
- However, this is **necessary** since we do not know anything about data outside of the support of P
- Need additional assumptions to do better
  - Focus on covariate shift



Image; Glauner et al., 2018

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Learning vs. Evaluation

- For this part, we will focus on **model evaluation**
  - **Learning:** Optimize $\mathbb{E}_Q[\ell(\theta; x, y)]$
  - **Evaluation:** Estimate $\mathbb{E}_Q[\ell(\theta; x, y)]$

- We will see why learning is harder later

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Integral Probability Metrics

- The **total variation distance** is

$$\mathrm{TV}(P, Q) = \int_{\mathcal{X} \times \mathcal{Y}} |q(x, y) - p(x, y)| \cdot dx \cdot dy$$

Sensitive to the mis-alignment between the two distributions. No notion of cost between two points.

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Integral Probability Metrics

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Sensitive to the mis-alignment between the two distributions. No notion of cost between two points.

- The **Wasserstein distance** is

$$W(P,Q) = \sup_{f:K_f \leq 1} \int_{\mathcal{X}\times\mathcal{Y}} f(x,y) \cdot (q(x,y) - p(x,y)) \cdot dx \cdot dy$$

$f(x,y)$: A smooth-function (Lipschitz constant less than 1) that measures how far apart the two distributions are. We are trying to find $f(x,y)$ that best separates the distributions.

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Evaluation Bounds

- Note that

$$\mathbb{E}_Q[\ell(\theta; x, y)] = \int_{X \times Y} \ell(\theta; x, y) \cdot q(x, y) \cdot dx \cdot dy$$

$$= \int_{X \times Y} \ell(\theta; x, y) \cdot (p(x, y) + q(x, y) - p(x, y)) \cdot dx \cdot dy$$

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$$= \int_{X \times Y} \ell(\theta; x, y) \cdot (p(x, y) + q(x, y) - p(x, y)) \cdot dx \cdot dy$$

$$= \int_{X \times Y} \ell(\theta; x, y) \cdot p(x, y) \cdot dx \cdot dy$$

$$+ \int_{X \times Y} \ell(\theta; x, y) \cdot (q(x, y) - p(x, y)) \cdot dx \cdot dy$$

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Evaluation Bounds

- Note that

$$\begin{aligned}
\mathbb{E}_Q[\ell(\theta; x, y)] &= \int_{X \times Y} \ell(\theta; x, y) \cdot q(x, y) \cdot dx \cdot dy \\
&= \int_{X \times Y} \ell(\theta; x, y) \cdot (p(x, y) + q(x, y) - p(x, y)) \cdot dx \cdot dy \\
&= \int_{X \times Y} \ell(\theta; x, y) \cdot p(x, y) \cdot dx \cdot dy \\
&\quad + \int_{X \times Y} \ell(\theta; x, y) \cdot (q(x, y) - p(x, y)) \cdot dx \cdot dy \\
&= \mathbb{E}_P[\ell(\theta; x, y)] + \int_{X \times Y} \ell(\theta; x, y) \cdot (q(x, y) - p(x, y)) \cdot dx \cdot dy
\end{aligned}$$

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Evaluation Bounds

- Note that

$$
\begin{aligned}
\mathbb{E}_Q[\ell(\theta;x,y)] &= \int_{\mathcal{X}\times\mathcal{Y}} \ell(\theta;x,y)\cdot q(x,y)\cdot dx\cdot dy \\
&= \int_{\mathcal{X}\times\mathcal{Y}} \ell(\theta;x,y)\cdot (p(x,y)+q(x,y)-p(x,y))\cdot dx\cdot dy \\
&= \int_{\mathcal{X}\times\mathcal{Y}} \ell(\theta;x,y)\cdot p(x,y)\cdot dx\cdot dy \\
&\quad + \int_{\mathcal{X}\times\mathcal{Y}} \ell(\theta;x,y)\cdot (q(x,y)-p(x,y))\cdot dx\cdot dy \\
&= \mathbb{E}_P[\ell(\theta;x,y)] + \int_{\mathcal{X}\times\mathcal{Y}} \ell(\theta;x,y)\cdot (q(x,y)-p(x,y))\cdot dx\cdot dy \\
&\le \mathbb{E}_P[\ell(\theta;x,y)] + \ell_{\max}\cdot \int_{\mathcal{X}\times\mathcal{Y}} |q(x,y)-p(x,y)|\cdot dx\cdot dy \\
&= \mathbb{E}_P[\ell(\theta;x,y)] + \ell_{\max}\cdot \mathrm{TV}(P,Q)
\end{aligned}
$$

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Evaluation Bounds

- Note that

$$
\begin{aligned}
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&= \int_{\mathcal{X}\times\mathcal{Y}} \ell(\theta; x, y) \cdot p(x, y) \cdot dx \cdot dy \\
&\quad + \int_{\mathcal{X}\times\mathcal{Y}} \ell(\theta; x, y) \cdot (q(x, y) - p(x, y)) \cdot dx \cdot dy \\
&= \mathbb{E}_P[\ell(\theta; x, y)] + \int_{\mathcal{X}\times\mathcal{Y}} \ell(\theta; x, y) \cdot (q(x, y) - p(x, y)) \cdot dx \cdot dy \\
&\leq \mathbb{E}_P[\ell(\theta; x, y)] + K_\ell \cdot W(P, Q)
\end{aligned}
$$

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Evaluation Bounds for Covariate Shift

- Note that

$$
\begin{aligned}
&\int_{\mathcal{X}\times\mathcal{Y}} \ell(\theta; x, y) \cdot (q(x, y) - p(x, y)) \cdot dx \cdot dy \\
&= \int_{\mathcal{X}\times\mathcal{Y}} \ell(\theta; x, y) \cdot (q(y \mid x)q(x) - p(y \mid x)p(x)) \cdot dx \cdot dy \\
&= \int_{\mathcal{X}\times\mathcal{Y}} \ell(\theta; x, y) \cdot (p(y \mid x)q(x) - p(y \mid x)p(x)) \cdot dx \cdot dy \\
&= \int_{\mathcal{X}\times\mathcal{Y}} \ell(\theta; x, y) \cdot p(y \mid x) \cdot (q(x) - p(x)) \cdot dx \cdot dy \\
&= \int_{\mathcal{X}} \left(\int_{\mathcal{Y}} \ell(\theta; x, y) \cdot p(y \mid x) \cdot dy \right) \cdot (q(x) - p(x)) \cdot dx \\
&= \int_{\mathcal{X}} \tilde{\ell}(\theta; x) \cdot (q(x) - p(x)) \cdot dx \\
&\leq K_{\tilde{\ell}} \cdot W(P(x), Q(x))
\end{aligned}
$$

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Evaluation Bounds for Covariate Shift

- Thus, we have

$$\mathbb{E}_Q[\ell(\theta; x, y)] \leq \mathbb{E}_P[\ell(\theta; x, y)] + K_{\tilde{\ell}} \cdot W(P(x), Q(x))$$

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Aside: What About Learning?

- Suppose we optimize the upper bound:

$$\mathbb{E}_Q[\ell(\theta; x, y)] \leq \mathbb{E}_P[\ell(\theta; x, y)] + K_{\tilde{\ell}} \cdot W(P(x), Q(x))$$

- It is equivalent to optimizing $\mathbb{E}_P[\ell(\theta; x, y)]$, since the penalty is independent of θ

- Recent approaches to use such bounds for learning

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Aside: What About Learning?

$$\mathbb{E}_Q[\ell(\theta; x, y)] \leq \mathbb{E}_P[\ell(\theta; x, y)] + K_{\tilde{\ell}} \cdot W(P(x), Q(x))$$

- Solutions for optimizing this upper bound (called Distributionally Robust Optimization):
  1. Formulate W() in the latent feature space, instead of raw input.
  2. Wasserstein Distributionally Robust Optimization: Assume Q to be within the vicinity of P and optimize loss considering that vicinity.
  3. Invariant Risk Minimization: Learn causal factors/variables/features that captures information/factors across different distributions & a linear classifier on top of such features generalizes better.